
Triple-Jumping over Fragmentation, Epidemics, and Spontaneous Adaption

Exploring Generating Function Methods in Adaptive Networks

By

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ABSTRACT

A network is a collection of nodes and edges, used ubiquitously to model real world systems. There exist network models that update node states in time, such as the voter model and the SIS model. When they also update edge configurations, they are called adaptive or co-evolutionary. These are both examples of network dynamics, and moment approximation schemes are employed to model these behaviours by counting small motifs. This report studies heterogeneous approximations, which are moments that incorporate node degrees. Dynamic systems of heterogeneous moments can be mapped to generating functions, producing PDEs.

For the SIS and adaptive voter model, the dynamics undergo distinct changes at parameter thresholds, and heterogeneous approximations can closely predict their location.

This report applies generating functions of heterogeneous approximations into three cases. First, the fragmentation threshold of the adaptive voter model is estimated using q -fan approximations in a generating function framework, giving analytic results for thresholds. Then, an error calculation is performed for the generating function of the active neighbourhood approximation for the SIS model, when one assumes a heterogeneous mean field ansatz. This shows which motifs are poorly estimated by the heterogeneous mean field. Finally, ways to construct single state networks with spontaneous dynamics from solvable PDEs are found using generating functions, in preparation for higher dimensional work.

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AUTHOR'S DECLARATION

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Research Degree Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, the work is the candidate's own work. Work done in collaboration with, or with the assistance of, others, is indicated as such. Any views expressed in the dissertation are those of the author.

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INTRODUCTION

In many disciplines, such as epidemiology and sociology, researchers use complex networks to model the interactions between sets of agents. These are often studied with numerical methods, but some of the most impressive insights have been reached analytically. Basic models studied in this format are the voter model, the SIS model, and their adaptive variants. These networks have complex dynamics, with analytic work requiring approximation schemes to capture their behaviour. Network behaviour may be simulated, but simulations do not directly reveal properties that characterise a model, such as bifurcations. Characteristic properties are instead found by using accurate approximation schemes and analytically solving for their dynamics. This information can be used to inform the design and modelling of complex systems. In this project, approximation methods for the dynamics of large complex networks are investigated, with the goal of developing entirely analytic approximations of dynamic network behaviour.

The report is structured as follows. In Chapter 1 we cover the important milestones in network science, up to the modern day, with the introduction of complex networks. In Chapter 2, the concepts and methods used in this report are outlined, where in Section 2.1 we introduce the voter model, adaptive voter model, and fragmentation. In 2.2 we introduce the SIS model and epidemic threshold. In Section 2.3 we introduce sets of approximation techniques for capturing the dynamics of agent-based models, methods to close the resulting equations, and discuss their dimensionality. These are split into homogeneous techniques, which give low-dimensional ODEs, and heterogeneous techniques, which give high dimensional ODEs. We then introduce more advanced approximation techniques, which give still higher dimensional ODEs, a procedure to convert approximations into low-dimensional PDEs, and the methods used to solve these PDEs. Chapter 3 goes onto applying generating function techniques on the adaptive voter model in the segregated state, using an active motif base. Chapter 4 then applies generating function techniques on the static SIS model, using active neighbourhood bases to analyse the accuracy of

the mean field approximation. Chapter 5 uses generating functions to create a general equation for spontaneous dynamics of a single state network, showing a relationship between dynamic equations and networks. Finally, in Chapter 6, this report concludes with a discussion of its content, possible future research, and the limitations encountered.

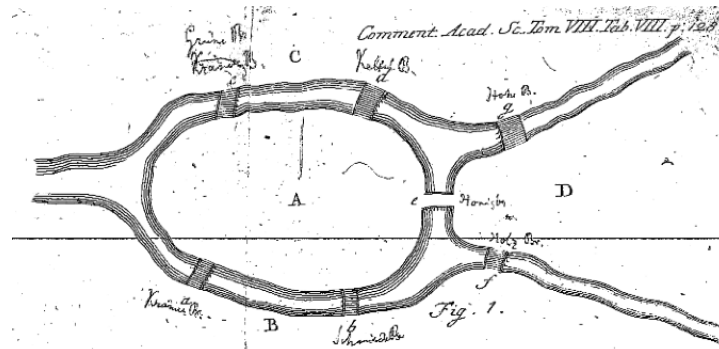


FIGURE 1.1. The Bridges of Königsberg, as they appear in Euler’s “Solutio Problematis ad Geometriam Situs Pertinentis”.

The network approach to modelling was first introduced by Euler [25], to model the bridges of Königsberg, and learn how to walk all the bridges without repetition. Landmasses were represented by nodes, and bridges were edges, creating the first network model. With this construction, Euler found the general condition that told him if he could walk all bridges only once. He could see, because of how many bridges each landmass in Königsberg had, that it was impossible. This discovery is the first in graph theory, the precursor to network science.

Networks have since been used to model animal territories [27], epidemics [61], opinion formation [73], and more. For example, Borůvka [12] minimised the amount of transmission line cable used in the Moravian power grid, by creating a minimum spanning tree. By representing roads as edges, and their intersections as nodes, we can use Dijkstra’s algorithm [21] to efficiently find a shortest path to a destination. Euler, Borůvka, and Dijkstra worked on topologically simple networks, constructed by simply placing nodes and assigning edge values.

The next major breakthrough in networks science came with the introduction of the Erdős-Rényi models [24, 26], $G(\mathcal{N}, M)$ and $G(\mathcal{N}, p)$. These have $\mathcal{N}$ nodes, assigning edges randomly. Only the $G(\mathcal{N}, p)$ has independent assignment, but is computationally complex, while the $G(\mathcal{N}, M)$ is computationally cheap.

The emergence of major links between the physics of complex networks, graph theory, biology, and sociology led to the birth of network science in the 1990s. This can be traced to Watts and Strogatz [74], on small world networks. Networks can also model is our social ties, and the small world phenomenon [48] of how pairs of total strangers can be linked by a short chain of acquaintances. Network science was then popularised by Barabási and Albert [6], on scale-free networks. These are networks that have a small number of interconnected “hubs”. In other

systems, these can model the structure of the internet. For example, how large sets of unconnected websites are all linked to a search website, and how search websites link to each other.

By now these milestones are the subject of major reviews and textbooks [1, 2, 7, 9, 22, 23, 50–52]. Current work aims to broaden the class of systems that can be described in the networks framework. The major fields are temporal networks [35], multilayer networks [31, 40], synchronisation phenomena [4], and adaptive networks [29].

Having been brought up to date on modern network science, the next chapter describes the models and methods used in this report. The models introduced are the adaptive voter model and the SIS model, while the methods are heterogeneous moment approximations and how to describe them with generating functions.

MODELS AND METHODS

We live in a global society with free access to information, and knowledge of how a population alters its opinions in time can have many uses. In recent news, companies such as Cambridge Analytica [13] have featured for exploiting knowledge of such dynamics to influence public opinions. The pieces of knowledge they use are the subject of much literature. The book, “Nudge” [68], discusses conditions for controlling group behaviour via behavioural economics, while the spread of misinformation is studied by Lewandowsky et al. [42] using psychology. How much one trusts a set of people can affect both these dynamics, with the tendency to trust like-minded individuals, known as homophily, playing into this. Extreme cases can result in a group isolating itself from outside perspectives, as shown by McPherson et al. [47] using human ecology, and by Colleoni et al. [16] using data analysis. To unify these results in a consistent framework, a number of network models describing opinion spread have been produced. Extended from the Ising model [36], these include the Sznajd model [67], the Schelling model [60], the voter model, and the adaptive voter model, which can describe isolationist scenarios through a fragmented absorbing state.

As ideas can spread, so can disease. Understanding how an epidemic is caused allows us to prevent it, and this knowledge can also be applied to the field of information infrastructure [58]. Epidemiology, which studies epidemics, is a multi-disciplinary subject introduced by Snow [65]. The spread of deadly outbreaks, such as the West African Ebola virus epidemic, are best contained by measures informed by epidemiological models [17]. As for opinion modelling, network models have been used to unify such disease spread into a consistent framework. The susceptible-infected (SI) model is the simplest, and describes diseases that cause permanent infectivity, such as HIV. With an added dynamic, the susceptible-infected-recovered (SIR) model [38] describes disease that can confer immunity, such as measles. This model has the advantage of having an exact analytic threshold for the incidence of an epidemic. Where the epidemic threshold is not exactly

known is the susceptible-infected-susceptible (SIS) model [3, 5, 33, 46], which describes diseases that allow repeat infection, such as the cold.

2.1 Voter Dynamics

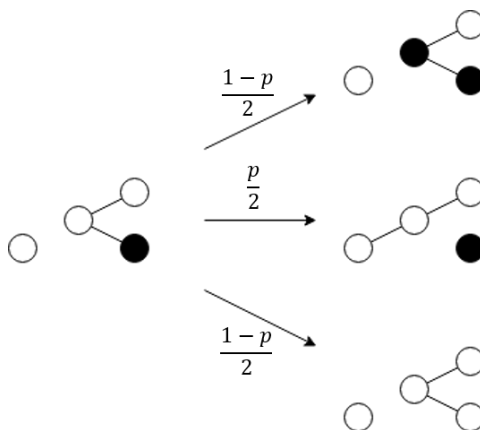


FIGURE 2.1. The agent-based update rules of the link update adaptive voter model, where p is the rewiring rate, and $1 - p$ is the adoption rate.

In the present project, a central role is played by the adaptive voter model. This is a type of model which exhibits dynamic temporal evolution, which is expressed by, in time, changing internal properties of nodes and the configuration of edges. The first is described by the dynamics *on* networks, and the second is described by the dynamics *of* networks. The section will now describe the various interpretations of the voter model and the adaptive voter model, and discuss the phenomenology of the adaptive voter model.

The voter model is the scheme from which the adaptive voter model was created. Introduced by Clifford and Sudbury [15] at the University of Bristol, and developed by Holley and Liggett [34], it is constructed as follows: each node represents a voter holding one of two opinions, and when an edge connects two voters of different opinion, one may adopt the opinion of the other. We call an edge like this *active*, and the major schemes by which this happens are link-update, forward-update, and backward-update.

The link-update voter model first selects an edge according to some point process. On an *active* edge, one node is randomly chosen, then for some p , with probability $1 - p$, they then synchronise to the state of the selected node, and the process is repeated. The forward voter model instead first selects a node and then randomly selects one of its neighbours. If the edge is active, then with probability $1 - p$, the neighbour is converted and the process is repeated. The backward voter model runs like the forward voter model, first selecting a node and then randomly selecting one of the edges connected to it. However, if the edge is active, then the neighbour converts the first node with probability $1 - p$, and the process is repeated.

There exist many other formulations of the voter model. A node may be able to select from more than two states, there may exist intermediary states, such that a node is undecided on what opinion it holds [56], or nodes may even take continuous values [32], which can have use in consensus modelling. The voter model only dynamically evolves the internal properties of nodes, which means it only exhibits dynamics *on* the network.

An important factor in real world communication is that people often prefer to communicate with others who share similar opinions, and the adaptive voter model accounts for this, as in Fig 2.1. The adaptive voter model [71] is such that on active edges, an adoption event may occur as in the voter model, but also a rewiring event. In rewiring, one of the nodes cuts the link, and then connects to a like-minded node instead. All major schemes of the voter model may be extended to include a rewiring procedure.

In this dissertation, we will be considering the link-update adaptive voter model, unless stated otherwise. It proceeds as follows. Edges are first randomly selected, according to a point process. On an active edge, one node is randomly chosen. With probability $1 - p$, nodes synchronise to the state of the selected node. However, with probability p , the selected node destroys the link and randomly reconnects to another node that shares its opinion. The process is then repeated.

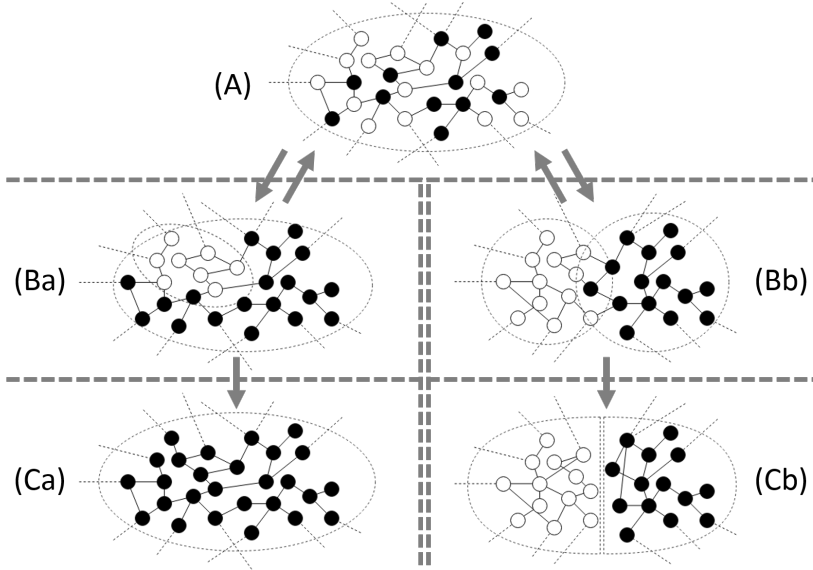


FIGURE 2.2. The possible states available for a binary adaptive voter model with rewiring rate p , where the transition point is p^* . (A) is the well mixed state, where n_a s and n_b s are intermingled. (Ba) is the connected segregated state before consensus, where the boundary region is small. (Bb) is the connected segregated state before fragmentation, where the boundary region is giant. (Ca) is consensus, occurring when $p < p^*$. (Cb) is fragmented, occurring when $p > p^*$. Arrows denote which states move into which other states.

The forward update adaptive voter model first selects a node and then randomly selects one

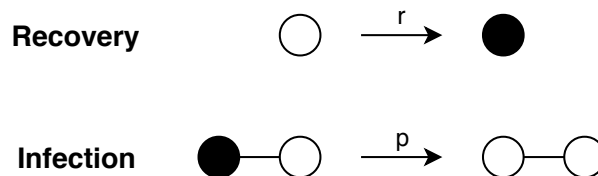


FIGURE 2.3. The two actions of the SIS model, where white nodes are infected, and black nodes are susceptible. The first action is recovery, while the second is infection.

of the edges connected to it. If the edge is activated, the first node either converts its neighbour with probability $1 - p$, or with probability p , the first node cuts the link, and rewires to a new like-minded node. The backward update adaptive voter model runs like the adaptive forward voter model, first selecting a node and then randomly selecting one of the edges connected to it. If the edge is activated, the neighbour either converts the first node with probability $1 - p$, but with probability p , the neighbour cuts the link, and rewires to a new like-minded node.

The forward and backward schemes make following a link more likely to lead to higher degree nodes, where the degree of a node is the number of endpoints of an edge connected to it. The link update scheme, however, gives either node on a selected edge identical statistical properties.

This construction means that the adaptive voter model adds dynamically evolving edge configuration, via the rewiring procedure, which is described by dynamics of the network. Rewiring also allows a network that is initially connected to turn disconnected. For most adaptive network models this *fragmentation* is not a problem as the dynamics can also reconnect disconnected components. However for the adaptive voter model, it is known [19, 71] that if the system rewires faster than some critical value, the *transition point*, then the system quickly approaches a fragmented absorbing state.

Before entering any absorbing state, the adaptive voter model enters a state of segregation, such that there are clusters that hold internally homogeneous but opposed opinions, with a very small border region of active links connecting them [60]. Then, at values above the transition point, the network splits into two giant components with mutually opposed but internally homogeneous states. This is fragmentation. Below the transition point, opinions intermingle until one opinion eventually dominates, creating an absorbing state with a single homogeneous giant component. This is an absorbing consensus state. The progression of these states is shown in Fig 2.2.

2.2 SIS Dynamics

Epidemic thresholds for the SIS model are also examined in this report. In the SIS model, nodes can be either susceptible S , or infected I . In random time, infected nodes can either infect susceptible nodes with a probability, p , or recover with a probability, r . Infection occurs along IS

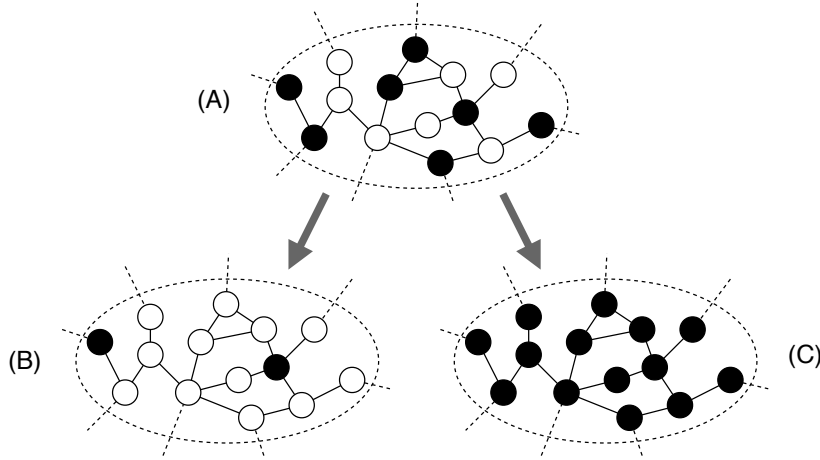


FIGURE 2.4. The three states for the SIS model, where black nodes are susceptible, and white nodes are infected. Infection occurs at a rate, p , and recovery at a rate, r . Call the epidemic threshold $p^*(r)$, a function of r . (A) is the well mixed state. (B) is the epidemic state, where $p > p^*(r)$. (C) is the cured state, where $p < p^*(r)$.

links, whereas recovery happens independent of a node's neighbours. This is distinct from the voter model in that the two states, S and I are not symmetric, however any I node becomes an S independently of its neighbours. They are together considered the two examples of minimal complexity, upon which naive assumptions often break down.

The two actions of infection and recovery thus have a nonlinear relationship regarding the long-term behaviour of an SIS model. With high enough p with respect to r , infected nodes may persist within the network for all time. This state is known as an epidemic. The alternative is that the infected nodes are eliminated, which we shall call the eradicated state.

Since recovery happens independent of a node's neighbours, it is always described in a node-update sense. Here, if a random node is selected, with probability, r , it becomes an S. Infection can be described, like in the voter model, in terms of a forward update, backward update, and link update procedure. An infection always occurs with probability, p . This report considers the forward update. For forward infection, from the set of nodes, if a random one is selected, and is infected, then it converts a random neighbour into an I. For backward infection, from the set of nodes, if a random one is selected, and is susceptible, then it adopts the state of a random neighbour. For link infection, from the set of links, if an SI or IS is selected, then they become an II link. This behaviour is shown in Fig 2.3 for any update type.

In the long term, the SIS model has two possible steady states, an epidemic (or endemic) or a healthy state. If the chance of infection, p , is high enough with respect to r , then a network will have an epidemic, where infected nodes persist in the network for all time. If p is low enough, the network is cured, such that the infection is eradicated and there are only susceptible nodes. The possible states for this are shown in Fig. 2.4.

The value at which p is large enough to cause an epidemic is a nonlinear function of r , denoted $p^*(r)$, where we drop the r . This value is called the *epidemic threshold*. The susceptible-infected-recovered (SIR) model acts similarly to the SIS. In this case, upon recovery, nodes become immune to the disease, and cannot be reinfected. Its construction means an analytic solution to the epidemic threshold has already been found when it was first constructed. However, no true analytic solution for the epidemic threshold of the SIS model has been found, but there are partial solutions [14, 69].

2.3 Mathematical Methods

Mathematical approaches have been proposed [10, 19, 30, 36, 39, 41, 44, 72] to describe the dynamics of agent-based network models. These use local update rules to create approximation schemes that provide such information in the form of a system of ordinary differential equations, and are called *moment expansions*.

The name originates from approximating a stochastic system by its moments, such as the mean, variance, skewness, etc. In network science, those we look at model the abundance of certain subgraphs or motifs in the network, where the number of links in a motif determines the order of the moment. This is called using a *motif basis*.

For general networks, either homogeneous or heterogeneous moment expansions are used to capture their dynamics, and require moment-closure techniques to close their equation systems. The homogeneous expansions use a motif basis that depends on the internal states of nodes, giving low-dimensional equation systems. These poorly capture the dynamics of the adaptive voter model. Heterogeneous expansions instead use a motif basis that also depends on the degree of the nodes. The dimensionality of the equation systems therefore scales with the maximum degree of any node. This means that in capturing more information, and generally providing better approximations for network dynamics, they result in high-dimensional equation systems. This report discusses both the homogeneous and heterogeneous mean field expansions and pair expansions. All provide poor estimates for the transition point of the adaptive voter model, but two methods provide partial solutions to this. Both use moment expansions with a motif basis that depends on given neighbourhood. The first uses the active motif expansion, which accurately predicts the transition point but needs to be solved numerically because of the high dimensionality [10]. The second uses the active neighbourhood expansion, which does not accurately predict the transition point, but uses a generating function mapping to reduce the high dimensional ODEs into low dimensional PDEs [63].

The first moment expansions, such as those of the Ising model [36], derived equations for single nodes of particular states, and are of zeroth order. Since these expansions only distinguish based on the internal states of nodes, they are homogeneous, and are known in the literature as homogeneous mean field expansions.

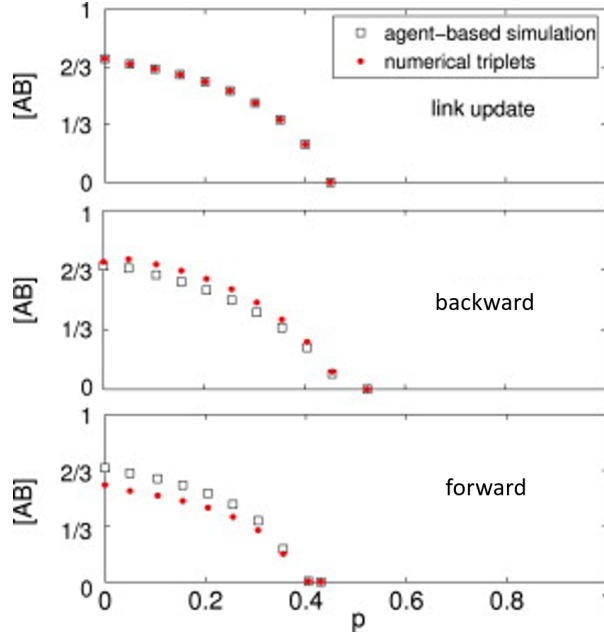


FIGURE 2.5. Simulations for $[n_a n_b]$ (written as $[AB]$ here) pairs as counted by the homogeneous pair expansion, for the link-update, forward, and backward adaptive voter models. Numerical results have been compared against agent-based simulations. Taken from Demirel et al. [19].

In general, we consider the link-update adaptive voter model, where nodes may take binary states a or b . Nodes in, say, state a are denoted n_a , and with scaling on our side, we think of p as the true rate parameter for any rewiring event, and $1 - p$ for any adoption event.

Here we show the homogeneous mean field expansion [19, 39, 72]. A common convention is to denote the density of a motif by wrapping $[]$ around the symbol for it, and another is to denote the rate of change for say, $[n_a]$, as $d[n_a]/dt = [\dot{n}_a]$. We then have that

$$[\dot{n}_a] = [n_a n_b] \left(\frac{1-p}{2} - \frac{1-p}{2} \right) = 0.$$

This is because the opinion update action is symmetrical along any link, and rewiring does not change internal properties of any nodes, treating the adaptive voter model as the voter model. What this means is, as a zeroth order expansion, the mean field expansion ignores all correlations, and assumes the distribution of node states is uniform.

This suggests using first order moments, so a candidate for accurate approximation is the homogeneous pair expansion, used previously to capture the dynamics of the adaptive susceptible-infected-susceptible (SIS) epidemic model [30]. Using the expansions from Demirel et al. [19], the dynamics of an $n_a n_a$ pair is given as an example, so that

$$[\dot{n}_a n_a] = \frac{1}{2} [n_a n_b] + \frac{1-p}{2} (2[n_a n_b n_a] - [n_a n_a n_b]).$$

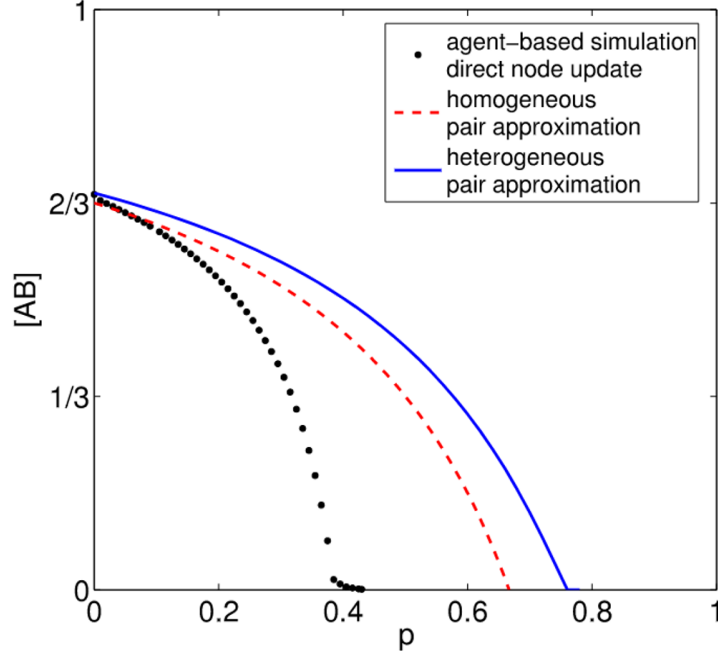


FIGURE 2.6. Performance of the heterogeneous pair approximation versus the homogeneous pair approximation and agent-based simulations in estimating the fragmentation threshold for the forward adaptive voter model. $N = 10^5$, $z = 4$. Taken from Demirel et al. [19].

Unlike the homogeneous mean field, this is not stationary. The first term looks at internal changes, which are the options that directly create or destroy $n_a n_a$ s, and their rates. The second term looks at external changes, that is, given some pair, what ways can an external node operate on this pair to create or destroy an $n_a n_a$. In each case, this creates some triplet, so rates of changes are dependant on triplets. This expansion approximates the dynamics of the adaptive voter model better than the mean field expansion, as demonstrated in Fig. 2.5 for the $n_a n_b$ pair.

The dynamics of $n_a n_b n_a$ and $n_a n_a n_b$ motifs are unknown without performing the second order homogeneous moment expansion, which would depend of motifs with 4 nodes, and so on. The systems of equations created by moment expansions are generally not closed, since they generally rely on moments of higher order than themselves. This means they rely on terms depending on motifs larger than those being estimated. A number of approximation techniques have hence been created to close the system, known as *moment-closure approximations* [19, 41].

The moment closure method mainly used in this project is a microscopic closure. This assumes weak correlations between motifs at the order of the expansion, so that it can be assumed that they occur independently of each other. The density of larger motifs can therefore be approximated as the products of the densities of smaller motifs. Two examples are given below for the moment-closure of the homogeneous mean field and homogeneous pair expansions.

Say the mean degree for nodes in an adaptive voter model is z . To approximate the number of

active pairs, $n_a n_b$, with single nodes, we note that an n_a must connect to an n_b . Assuming single nodes occur independently means that, for a given link, the probability it connects an n_a and n_b is simply $[n_a][n_b]$. Using the mean degree, we assume any node is connected to z more. This lets us say that the density of any active link,

$$[n_a n_b] = z[n_a][n_b].$$

In general, this is for closing a homogeneous mean field expansion, however it has been shown that for the adaptive voter model this gives stationary dynamics. We consider then the homogeneous pair approximation.

The excess degree of a node is defined as the number of extra neighbours a node has after arriving to it from some edge. Say now that the mean excess degree is g . To close the homogeneous pair expansion, the triplets, $n_a n_b n_a$ and $n_a n_a n_b$ must be reduced. This would give us

$$(2.1) \quad [n_a n_b n_a] = \frac{g[n_a n_b]^2}{2z[n_b]},$$

which is explained in the following. The idea is as above, using $n_a n_b$ s to make an $n_a n_b n_a$, but addressing all double counting that a multiplication would cause.

To create an $n_a n_b n_a$, two $n_a n_b$ s must occur. But counting both leads to double counting the n_b . Thus we must divide through by its density. An $n_a n_b n_a$ also occurs when an $n_a n_b$ and a single n_a is reached following the n_b . Since following the link in the pair leads us to a node with g many further connections, we multiply through by g . But we are now double counting pairs of links, too, since this works in both directions. The mean degree is z , so we divide through, removing the second direction. Finally, we note that any $n_a n_b n_a$ is symmetric, so we divide through by 2. This gives Equation (2.1).

When initiating on an Erdős-Rényi random graph, the mean degree and mean excess degree are identical, so $g/z = 1$, but this is only correct assuming stationary distributions. The adaptive voter model was found to, despite rewiring, remain almost perfectly Poissonian, and so the approximation has yielded good results [30].

For the adaptive voter model, these moment-closure approximations, applied to the homogeneous pair expansion, overestimate the rewiring rate at which fragmentation occurs [19]. Higher order homogeneous motifs have therefore been tested, and they tend to lead to better approximations, as in for the adaptive SIS model [37, 43, 75]. However, for the adaptive voter model, as higher order expansions are used, convergence to a correct prediction of the transition point is neither quick nor uniform [19]. In addition, as order increases, the ODEs quickly become more complex. This may suggest alternative closures must be applied to accurately capture the dynamics of the adaptive voter model.

Alternatives to the above moment closures have also been suggested, having been avoided by generating their effects through numerical simulation [28], or by assuming maximum-entropy to generate closure schemes [57]. The first generates results for the adaptive SIS, and the second

does so for the SIR. These two methods also avoid many network structure assumptions, but neither is fully analytic. This suggests that expansions that capture more diverse information sets are necessary. One possible solution is a *heterogeneous expansion* that captures the degree of each node in the motifs, as well as the node state.

The trade-off for using heterogeneous expansions is that the dimensionality of the equation system scales with the maximum degree of any node, $d_{\max}$. For the general N state adaptive voter model, this is shown in Table 2.1, which lists the dimensions associated with homogeneous, heterogeneous, and more complex expansions.

Expansion	ODE Dimension	PDE Dimension	Space Dimension
Homog. Mean Field	N	—	—
Homog. Pair	$N + \frac{N(N+1)}{2}$	—	—
Hetero. Mean Field	$Nd_{\max}$	N	1
Hetero. Pair	$\left(\frac{N(N+1)}{2}\right)d_{\max}^2 + Nd_{\max}$	$N + \frac{N(N+1)}{2}$	2
Active Nbhd.	$N^2 \left(\binom{d_{\max}+N}{N} - 1\right)$	N	N
Active Motif	To Be Completed	To Be Completed	To Be Completed

TABLE 2.1. The dimensionality of the ODE, PDE, and PDE space of the equation systems associated with particular expansions for an adaptive voter model with N states and maximum degree $d_{\max}$. The PDE and Space columns refer to the resulting dimensionality after a generating function mapping is applied to the ODE system of the expansion. This is in line with the triple jump method [63].

Of the heterogeneous expansions, the simplest is the heterogeneous mean field expansion [54], which is of the zeroth order, and then the heterogeneous pair expansion [45], which is of the first order, performed by Demirel et al. [19] for the forward adaptive voter model. The heterogeneous pair expansion actually performed worse than the homogeneous pair expansion in approximating the number of active motifs, shown in Fig. 2.6. This suggests heterogeneous expansions of this type still have insufficient information.

We now introduce three moment expansion schemes that consider the neighbourhoods of nodes, not just the degree. The first is the active neighbourhood expansion. We use the active neighbourhood in the context of the SIS model, but present it here in the form of the adaptive voter, as in Silk et al. [63].

The adaptive voter model is such that a node is only more active with neighbours of opposite opinion. More neighbours of similar opinion suggests deeper embedding in a segregated cluster, and so would correlate with a slower rate of change. An expansion scheme has therefore been developed, where the motif base depends on neighbour state. Used by Silk et al. [63] to illustrate an analytic method of moment expansion, the *active neighbourhood expansion* [44] does not consider nodes of given degree, rather nodes with a given neighbourhood.

Denote an a state focal node as an $a_{k,l}$, where k is the number of a neighbours, and l is the number of b neighbours. For the link update adaptive voter model [63], the moment expansion

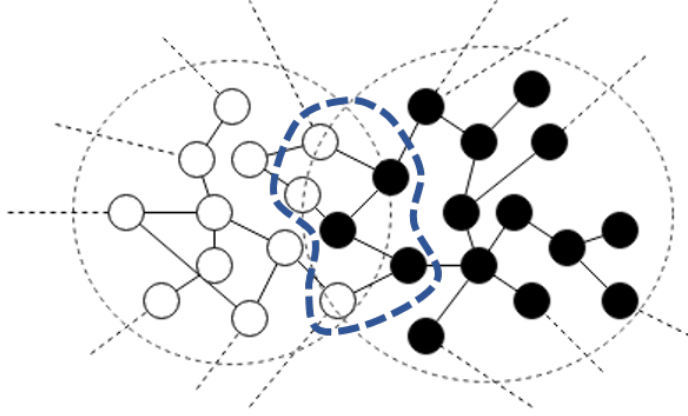


FIGURE 2.7. A binary adaptive voter model in a state of connected segregation, where the border nodes are wrapped by the blue dotted line.

for some $a_{k,l}$ is then as in (2.2).

$$\begin{aligned}
 (2.2) \quad [a_{k,l}] &= \frac{1-p}{2} (k [b_{k,l}] - l [a_{k,l}]) \\
 &+ \frac{1-p}{2} ((l+1) [a_{k-1,l+1}] - l [a_{k,l}]) \\
 &+ \frac{1-p}{2} \frac{\sum_{k,l} (k-1)k [b_{k,l}]}{\sum_{k,l} k [b_{k,l}]} ((l+1) [a_{k-1,l+1}] - l [a_{k,l}]) \\
 &+ \frac{1-p}{2} \frac{\sum_{k,l} lk [a_{k,l}]}{\sum_{k,l} k [a_{k,l}]} ((k+1) [a_{k+1,l-1}] - k [a_{k,l}]) \\
 &+ \frac{p}{2} ((l+1) [a_{k-1,l+1}] - l [a_{k,l}]) \\
 &+ \frac{p}{2} ((l+1) [a_{k,l+1}] - l [a_{k,l}]) \\
 &+ \frac{p}{2} \frac{\sum_{k,l} l [a_{k,l}]}{\sum_{k,l} [a_{k,l}]} ([a_{k-1,l}] - [a_{k,l}]),
 \end{aligned}$$

where the terms in the summations are all microscopic closure approximations of higher moments. An analogous expansion exists for b focal nodes, which are denoted $b_{k,l}$. This approximation functions well for small p , but as p increases, it diverges further from results obtained by simulation [19]. The transition point is therefore poorly estimated by the active neighbourhood expansion. This is likely because the active neighbourhood expansion is of the zeroth order, so does not consider correlations beyond nearest neighbour interactions, and suggests that the expansion scheme must be extended.

We now discuss a method that accurately predicts the transition point, at the expense of needing to be solved numerically because of its high dimensionality. To actually capture the dynamics of the adaptive voter model, one would need to consider all motifs with active links. The *active motif approach* suggests that only considering these is necessary, disregarding inactive

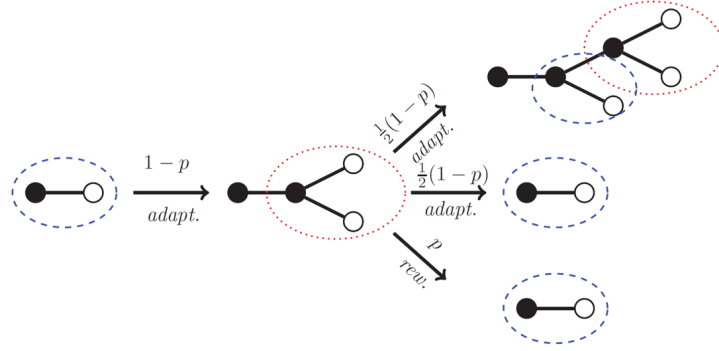


FIGURE 2.8. Update rules for the 3-regular active motif approximation, taken from Böhme and Gross [10].

motifs, but this is still a large set. It is however reasoned that only motifs of an order up to mean degree need be considered [19]. For highly connected graphs, this remains very difficult. But to predict the transition point, we only need to examine the space of network configurations before fragmentation, which is always a connected segregation state. In segregation, all active motifs share space on the border region, and only this part of the motifs is active, as in Fig 2.7. Using a bootstrapping argument that says the dynamics of an active motif depend entirely on its section on the border region, we may safely use the following expansion.

The active motifs considered here are made up of a *focal node* and the neighbours that form active links, called *fringe nodes*. One formulation was used to successfully predict the transition point of the adaptive voter model [10], and another for the multi-state adaptive voter [11].

The advantage of only using motifs on the border region is that it is much smaller than the whole network. Therefore, active motifs have a very low density on the network, so that the expansion gives a system of linear equations. The time evolution about the segregated state is thus fully captured by the eigenvalues of the corresponding Jacobian matrix, allowing the use of bifurcation analysis.

If the eigenvalues are all negative, then the low density (fragmented) state of active motifs is stable and declining, meaning the network will fragment. The converse means the fragmented state is unstable, so the network will then remain connected and the active links will propagate.

For simplicity, the first scheme [10] considers a degree-regular network, so that all nodes have the same degree k . This scheme considers q -fan motifs, such that about a given focal node, there exist q active links.

Following [10] we now describe the expansion for $k = 3$. Assuming a state of segregation, begin with only one active link, and so initiate on a 1-fan. The focal node may rewired, removing a 1-fan, since the fringe node and focal node are now only connected to similar nodes. However, either node may adopt, destroying a 1-fan, and turning either node into the focal node of a new 2-fan, as they initially had two similar neighbours.

The expansion proceeds for the 2-fan. From here, there are three possible events: any fringe

node may adopt, spawning a 2-fan and an added 1-fan; or the focal node may adopt, removing the 2-fan to return a 1-fan; or any node may rewire, again removing the 2-fan and returning a 1-fan. This is visualised with rates in Fig 2.8.

For any k -regular graph in the segregated state, an active k -fan is an isolated node. Due to the segregation assumption, if a node is isolated then it is the only node of its type and is an instance of a small boundary region. Since $[k] = 1$, and does not grow with the network, its proportion of the network approaches zero at the same rate as the network's growth, so is not considered. Due to rewiring only increasing the number of same neighbours, no $(k + 1)$ -fans can be created. All fringe nodes are only connected to one similar neighbour. These properties together naturally function as an approximation free moment closure, such that only $(k - 1)$ -fans and below are active. The actions and their rates are summarised in (2.3) and (2.4), such that for any k -regular network,

$$(2.3) \quad 1\text{-fan} \xrightarrow{1-p} (k-1)\text{-fan},$$

and for $q \in \{2, \dots, k-1\}$,

$$(2.4) \quad q\text{-fan} \rightarrow \begin{cases} \xrightarrow{(1-p)/2} 1\text{-fan}, & \text{Base Adoption;} \\ \xrightarrow{(1-p)/2} (q-1)\text{-fan} + (k-1)\text{-fan}, & \text{Fringe Adoption;} \\ \xrightarrow{p/2} (q-1)\text{-fan}, & \text{Base Rewiring;} \\ \xrightarrow{p/2} (q-1)\text{-fan}, & \text{Fringe Rewiring.} \end{cases}$$

Counting the number of motifs in the network, the corresponding rates of change are then

$$(2.5) \quad \begin{aligned} [\dot{1}] &= -[1] + 2[2] + \frac{1-p}{2} \sum_{g=3}^{k-1} g[g]; \\ [\dot{q}] &= -q[q] + \frac{1+p}{2} (q+1)[q+1], & q \in \{1, \dots, k-2\}; \\ [k-1] &= (1-p)[1] - (k-1)[k-1] + \frac{1-p}{2} \sum_{g=2}^{k-1} g[g], \end{aligned}$$

where $[0]$ is defined as zero. This scheme improves upon previous ones, closely predicting the transition point [10], since the adaptive voter model has a very narrow degree distribution. However, the degree-regular assumption is untrue after the first rewiring. The scheme also does not consider inert links. This means that the motif base has no measure indicating how embedded a node is within a similar cluster. Therefore, a second scheme for general degree distributions was also created [10], which counts (m, l) spiders, such that about a given focal node, there are m inert links and l active links. The degree of the focal node is therefore $d = m + l$. This is similar to the active neighbourhood approximation, but as a type of active motif approach, only spiders with $l > 0$ are considered. In addition, the spiders are defined by number of active neighbours and inactive neighbours, not base node. For any (m, l) spider, there are four possible distinct events.

If rewiring occurs, either the focal node keeps the link, returning an $(m+1, l-1)$ spider, or a fringe node keeps the link, returning an $(m, l-1)$ spider. If adoption occurs, either the focal node adopts, returning an (l, m) spider, or the fringe node adopts, returning an $(m+1, l-1)$ spider and creating a new $(1, g)$ spider.

The new spider likely has only 1 similar neighbour due to the small size of the border region, and the rest of its neighbours are embedded in the cluster it was originally part of. Since we follow a random edge to reach the focal node of the new spider, g is selected according to the excess degree distribution. This is visualised with rates in Fig 2.9, and the general transitions are shown in Equation (2.6), where the maximum degree, d , of any node is denoted as $d_{\max}$.

$$(2.6) \quad (m, l) \rightarrow \begin{cases} \xrightarrow{(1-p)/2} (l, m), & \text{Base Adoption;} \\ \xrightarrow{(1-p)/2} (m+1, l-1) + \sum_{g=1}^{d_{\max}-1} \mathbb{P}\{g+1\}(1, g), & \text{Fringe Adoption;} \\ \xrightarrow{p/2} (m+1, l-1), & \text{Base Rewiring;} \\ \xrightarrow{p/2} (m, l-1), & \text{Fringe Rewiring.} \end{cases}$$

Assume that this takes places on an Erdős-Rényi random graph. With this assumption, [19] shows that the rate equations for any active (m, l) spider is

$$(2.7) \quad \begin{aligned} \dot{m}[l] = & -l[m, l] + \frac{1-p}{2}m[l, m] + \frac{p}{2}(l+1)[m, l+1] \\ & + \mathbb{1}_{m \neq 1}(m) \frac{1}{2}(l+1)[m-1, l+1] + \mathbb{1}_{m=1}(m) \frac{1-p}{2} \underbrace{\frac{e^{\langle d \rangle} \langle d \rangle^{l+1}}{(1+l)!}}_{(2.7a)} \sum_{[u, v]} v[u, v] \end{aligned}$$

where $\mathbb{1}$ is the indicator function, and $\langle d \rangle$ is the mean degree of the network. Since this is assumed to be an Erdős-Rényi graph, the degree distribution is Poissonian, which is what gives (2.7a). The Jacobian for the system of equations generated by this procedure is of dimensions $D \times D$, where

$$(2.8) \quad D = \frac{d_{\max}(d_{\max} + 1)}{2}.$$

This means that as $d_{\max}$ increases, so does the Jacobian, becoming very difficult or impossible to solve, and so is limited in use for general degree distributions. Without analytic procedures, this necessitates numeric computation to find the transition point. Böhme and Gross [10] only simulated spiders up to $d_{\max} = 50$, as they were limited by compute power and time. This shows that methods to solve the moment-expansions analytically are necessary to study larger networks with higher degrees.

The analytic method this project uses is the three step, *mathematical triple-jump*, introduced by Silk et al. [63], which reduces high-dimensional ODEs into low-dimensional PDEs. The first step is to initiate some moment expansion and moment-closure approximation on an agent-based model, to capture its dynamics. The second is to map the terms in the resultant evolution equations to generating functions [66], which turns the high-dimensional ODE system into

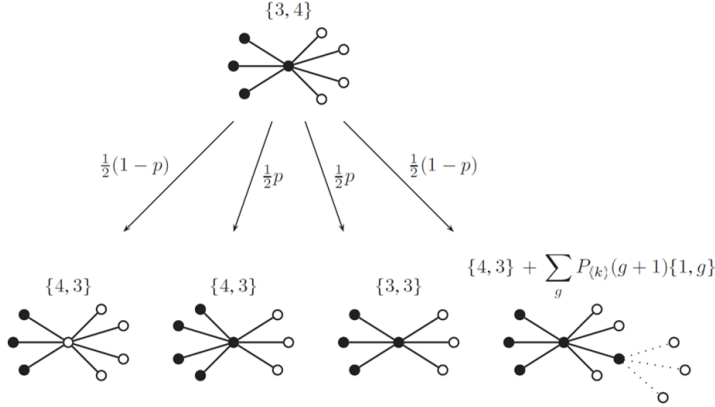


Figure 2.9: Update rules for a case of the general degree distributed active motif approximation, taken from Böhme and Gross [10]. Here, $P_{\{k\}}$ denotes probability of a degree, given connection to a node of degree k .

low-dimensional PDE system. The final step is to then solve the PDE system, using suitable techniques.

This method has been used successfully to model food chain networks [8], to design a method to create conditions for self-organisation [64], and to analyse epidemics [20]. Similar methodology has also been used by other authors [49, 55].

Silk et al. [63] performed the triple jump on the active neighbourhood expansion, which has already been described, giving (2.2). This gives a pair of infinite-dimensional ODE systems, and generating functions [66] can now be used to reduce them to a pair of two-dimensional PDEs, where this is the second step. The original paper uses two schemes, where the first assumed $l[a_{kl}] = k[b_{kl}]$, and the second removed this assumption. Each have $a_{k,l}$ and $b_{k,l}$ interpreted as the coefficients of Taylor style polynomials, giving

$$Q(t, x, y) = \sum_{k,l} a_{k,l}(t) x^k y^l; \quad R(t, x, y) = \sum_{k,l} b_{k,l}(t) x^k y^l,$$

such that x and y are indexing parameters, and Q and R are generating functions. These mappings are bijective, so solving the PDE system for Q and R gives a solution that, in this case, captures the dynamics of the adaptive voter model, which forms the third and final step. The method to solve the generating functions depends on the structure of the PDE, but all involve standard generating function techniques, this instance was produces a first-order scalar quasilinear equation, which can be solved using the method of characteristics [53].

More generally, for the second case when $l[a_{k,l}] \neq k[b_{k,l}]$, a second-order system in two dimensional space is generated which has not yet been solved, but the existence of an analytic solution may be possible. If this were solved, then the active neighbourhood expansion would be fully described for the two-state adaptive voter.

For an adaptive network in three states, performing the triple jump procedure on an active neighbourhood expansion would result in a three-dimensional PDE in three-dimensional space.

As in Table 2.1, the general active neighbourhood for the N -state adaptive voter model has an ODE dimensionality of

$$N^2 \left(\binom{d_{\max} + N}{N} - 1 \right),$$

because it is an unordered sampling with replacement from a set of size N , from 1 to $d_{\max}$ many times, with a central node that can take N states. Converting the general active neighbourhood expansion into a PDE system then gives an N -dimensional PDE in N -dimensional space, and the resulting equations have not been solved.

Generally speaking, the method used to solve a PDE system is defined by its class, is a consequence of its order and characteristics. For some function, $u(x_1, x_2, \dots)$, order is defined by the largest number of differentiations performed upon u . Denote derivatives as $\partial u / \partial x_i = u_{x_i}$. For some function, $\mathbf{F}$, a first order system of the general form $\mathbf{F}(x_1, x_2, \dots, u, u_{x_1}, u_{x_2}, \dots)$, a solution can always be found using the method of characteristics.

This is not generally the case for second order systems. Depending on whether they are linear or nonlinear, solutions can be found in different ways. The effects of the structure on the solvability of linear PDEs is discussed in Section 6. Nonlinear PDEs of the second order and greater, where the nonlinearity can also be expressed in multiple function PDEs, are typically unsolvable. In the nonlinear case, the sum of the degrees of each irreducible function denotes the PDE's dimension, D_p , and the number of independent parameters denotes the dimension of the space it occupies, D_s . If $D_s > D_p$ or $D_s = D_p = 1$, then the PDE is solvable. This is because it can be reduced by a change of variables into linear form. However, when $D_p \geq D_s > 1$, the set of equations are not generally solvable.

A central challenge is then knowing if PDEs resulting from the space of moment expansions can be solved. The triple-jump procedure reveals a pattern, as in Table 2.1. Higher order expansions will likely have higher dimensionality, and a number of factors in the choice of motif basis seem to directly affect the dimensionality of the resultant PDE and its space. These include motif shape, order, number of possible node states, and heterogeneity. For the adaptive voter model, the triple jump procedure has not been applied beyond the active neighbourhood, such that the dimensionality of the active motif expansion is not known, as stated in Table 2.1.

This chapter has described the voter model, adaptive voter model, the SIS model, and transition points. Homogeneous and heterogeneous moment approximations, which are methods for capturing the dynamics of agent-based models, have been detailed. Of the heterogeneous expansion, active motif methods were the focus, specifically detailing q -fans and (m, l) spiders. The active neighbourhood expansion, which also counts inactive motifs, was also described. Generating function procedures were introduced in the context of Silk et al. [63], with the aim of studying them as PDEs, in the *triple-jump* procedure. The bootstrapping arguments of Böhme and Gross [10] were also described, and the last part discussed the effects of dimensionality on PDEs, and which are expected to be solvable or not.

The following three chapters detail the results obtained during the research in this report. Chapter 3 builds on the work of Böhme and Gross [10] for the fragmentation transition of the adaptive voter model. Here we apply the triple jump procedure onto the system of equations, and then analyse the stability of the resulting PDE. This has the purpose of giving an accurate analytic prediction of the fragmentation transitions for graphs with any degree distribution. Chapter 4 builds on the work of Silk [62] for the epidemic threshold of the SIS model. Here we apply generating function properties to find sources of inaccuracy in the mean field approximation for predicting the epidemic threshold, by assuming it to be correct for the active-neighbourhood approximation. By doing so, this informs further analysis into analytic calculations of the threshold. Chapter 5 uses triple jump methodology to look at how to produce network models from given solvable PDEs, by creating a general equation to govern spontaneous dynamics of a single state network. This provides a minimal working example of how to characterise the space of solvable network models.

EVALUATING q -FANS

This chapter develops analytic solutions for the fragmentation transition of the adaptive voter model, which are all presently inaccurate or limited in some way. In [10], a q -fan approximation and stability analysis of the Jacobian gave accurate fragmentation estimates for networks with low mean degree, and underestimates for high mean degree. This inaccuracy is likely caused by having assumed degree regularity, since rewiring always has the network approach a Poissonian distribution. In addition, the size of the Jacobian scales with maximum degree, as in (2.8). This becomes analytically intractable and gives little insight into the properties of the model. Alternatively, in Silk et al. [63], the dynamics of the adaptive voter were captured with an active neighbourhood approximation and analysed with generating functions. This accurately estimated the fragmentation transition for high mean degree networks, but overestimated for low mean degree. The generating function procedure can be employed when capturing with q -fans, which is the subject of this chapter.

This section applies the triple-jump framework to the q -fans at segregation, assuming regularity, as a step on to analysing broad degree distributed networks. Doing so with matrix analysis requires a cutoff, else one obtains an infinite matrix. However, generating functions enumerate the degree distributions of each fan into one infinite dimensional equation. Without such limitations, fan dynamics may be selected from a distribution, rather than deterministically.

We proceed by evaluating the dynamics of the q -fan motif approximation for a binary adaptive voter model of rate p , for a general k regular graph in a segregated state. In Chapter 2, (2.3) and (2.4) were given, describing the rates of change of each active fan motif on a k -regular graph. Using these, we proceed with the triple jump procedure. Define the generating function tracking active fan motifs on this as

$$G(t, x; p) = \sum_q [q](t; p) x^q,$$

where $[q]$ is the approximate number of active q -fans in a large network with rate p at time t ,

and x is an arbitrary spacial indexing variable. Approximations are up to the network order, so that each $[q]$ is giant. Unless unclear, we drop the p and t dependencies, and denote $\partial G/\partial t = G_t$.

Initialisation on the segregated state of any k regular graph means that for all $j \geq k$, and for all time, $[j] = 0$, and $[0] = 0$. The first is because rewiring almost surely connects to embedded nodes, which in finite time almost surely never become active, while the second is an active motif assumption. This means that $G(t, x) = \sum_{q=1}^{k-1} [q]x^q$. From this, it can be seen that

$$G_t(x) = \sum_{q=1}^{k-1} [\dot{q}]x^q = [\dot{1}]x + \sum_{q=2}^{k-2} [\dot{q}]x^q + [k-1]x^{k-1}.$$

Next is to expand each term to obtain sums from $i = 1$ to $k-1$. Expanding the first term gives

$$\begin{aligned} (3.1) \quad [\dot{1}]x &= -[1]x + 2[2]x + \frac{1}{2}(1-p) \sum_{q=3}^{k-1} q[q]x \\ &= -[1]x + 2[2]x + \frac{1-p}{2} \left(\sum_{q=1}^{k-1} q[q]x - 2[2]x - [1]x \right) \\ &= \frac{p-3}{2}[1]x + (1+p)[2]x + \frac{1-p}{2} \sum_{q=1}^{k-1} q[q]x. \end{aligned}$$

Following on, expanding the second term gives

$$\begin{aligned} (3.2) \quad \sum_{q=2}^{k-2} [\dot{q}]x^q &= \sum_{q=2}^{k-2} -q[q]x^q + \frac{1}{2}(1+p)(q+1)[q+1]x^q \\ &= \sum_{q=1}^{k-1} -q[q]x^q + \frac{1+p}{2}(q+1)[q+1]x^q \\ &\quad - \left(-[1]x + \frac{1+p}{2}2[2]x \right) - \left(-(k-1)[k-1]x^{k-1} + \underbrace{\frac{1+p}{2}k[k]x^{k-1}}_{=0} \right) \\ &= \sum_{q=1}^{k-1} \left(\frac{1+p}{2}(q+1)[q+1] - q[q] \right) x^q + ([1] - (1+p)[2])x + (k-1)[k-1]x^{k-1}. \end{aligned}$$

Lastly, expanding the third term gives

$$\begin{aligned} (3.3) \quad [k-1]x^{k-1} &= -(k-1)[k-1]x^{k-1} + (1-p)[1]x^{k-1} + \frac{1}{2}(1-p) \sum_{q=2}^{k-1} q[q]x^{k-1} \\ &= -(k-1)[k-1]x^{k-1} + (1-p)[1]x^{k-1} + \frac{1-p}{2} \left(\sum_{q=1}^{k-1} q[q]x^{k-1} - [1]x^{k-1} \right) \\ &= -(k-1)[k-1]x^{k-1} + \frac{1-p}{2}[1]x^{k-1} + \frac{1-p}{2} \sum_{q=1}^{k-1} q[q]x^{k-1}. \end{aligned}$$

Summing (3.1), (3.2), and (3.3) through then gives

$$(3.4) \quad G_t(x) = \frac{1-p}{2} (x^{k-1} - x) [1] + \frac{1-p}{2} (x + x^{k-1}) \sum_{q=1}^{k-1} q[q] - \sum_{q=1}^{k-1} q[q]x^q + \frac{1+p}{2} \sum_{q=1}^{k-1} (q+1)[q+1]x^q.$$

Next (3.4) is rewritten in terms of G and its partial derivatives. It can be seen that

$$(3.5) \quad [1] = G_x(0) = G_{x,0}; \quad \sum_{q=1}^{k-1} q[q] = \sum_q q[q] = G_x(t, 1) = G_{x,1},$$

such that each is independent of x . By factorising x , and differentiating, the solution is

$$\sum_{q=1}^{k-1} q[q]x^q = x \sum_{q=1}^{k-1} q[q]x^{q-1} = x \frac{\partial}{\partial x} \sum_{q=1}^{k-1} [q]x^q = x \frac{\partial}{\partial x} \left(\sum_q [q]x^q \right) = x G_x(x).$$

The last term is solved by shifting indices, cancelling out $[1]$, and then differentiating, such that

$$\sum_{q=1}^{k-1} (q+1)[q+1]x^q = \sum_{q=2}^{k-1} q[q]x^{q-1} = \sum_{q=1}^{k-1} q[q]x^{q-1} - [1] = \frac{\partial}{\partial x} \left(\sum_{q=1}^{k-1} [q]x^q \right) - G_{x,0} = G_x(x) - G_{x,0}.$$

Thus $G_t(x)$ can be rewritten as

$$(3.6) \quad G_t(x) = \frac{1-p}{2} (x^{k-1} - x) G_{x,0} + \frac{1-p}{2} (x + x^{k-1}) G_x(1) - x G_x(x) + \frac{1+p}{2} (G_x(t, x) - G_{x,0}),$$

which is

$$(3.7) \quad G_t(x; p) = \left(\frac{1-p}{2} (x^{k-1} - x) - \frac{1+p}{2} \right) G_{x,0} + \frac{1-p}{2} (x + x^{k-1}) G_{x,1} + \left(\frac{1+p}{2} - x \right) G_x(x).$$

Now consider when the network is in the fragmented state. As shown in Fig. 2.2, this is a stationary state, so $G_t(x) = 0$. Assume that the rate of change of any motif is continuous, so $G(x)$ is well behaved. Since $G_t(t, x) = 0$ then for some function G^* , $G(t, x; p) = G^*(x; p)$, where G^* has no time dependence. Furthermore, there are no active motifs in the fragmented state and so there are no 1-fans or greater. Since each $[q] \geq 0$ for all possible q , this all implies that in the fragmented state, (3.5) means,

$$0 = \left(\frac{1+p}{2} - x \right) G_x^*(x).$$

This gives two solutions, where either

$$x = \frac{1+p}{2} \quad \text{or} \quad G^*(x) = \text{constant}.$$

The x solution is only true in the fragmented state, not for all time, expressed by how (3.7) is nonautonomous. Since $G^*(0) = 0$, we have that $G^*(x) = 0$, and so $G^*(t, x) = 0$ is the trivial solution.

Consider instead if (3.7) is stationary while active motifs still exist. This is an equilibrium state where for all q , each $[q](t)$ is stable with respect to time, but not necessarily zero, so not fixed. However, the generating function itself moves from an unstable configuration to a stable configuration, and this transition point will provide the estimate for the fragmentation transition. This is analogous to the stability analysis of the Jacobian in Böhme and Gross [10], and is finding the same values. The difference is that, when a Jacobian is a large multidimensional object when eigenvalues are challenging to find, the generating function is one dimensional, and finding eigenfunctions is straightforward for arbitrary k . To find eigenfunctions, the objective equation is

$$(3.8) \quad \lambda G(t, x; p) = G_t(t, x; p).$$

We shall drop the t since this will be a solution in the steady state, and drop p until it is explicit. Inserting this into (3.7) then gives, after rearranging,

$$(3.9) \quad G'(x) + \frac{2\lambda}{(2x - (1 + p))} G(x) = \frac{((1 - p)(x^{k-1} - x) - (1 + p))}{(2x - (1 + p))} G'(0) + \frac{(1 - p)(x + x^{k-1})}{(2x - (1 + p))} G'(1).$$

This is of the form $y' + f(x)y = g(x)$, which is a first order linear ODE, solved using integrating factors. This equation uses

$$(3.10) \quad f(x) = \exp\left(\int \frac{2\lambda}{(2x - (1 + p))} dx\right) = \exp(\lambda \ln |2x - (1 + p)|) = \exp(\ln |2x - (1 + p)|)^\lambda$$

such that

$$(3.11) \quad f(x) = \begin{cases} ((1 + p) - 2x)^\lambda, & x \leq \frac{1+p}{2}; \\ (2x - (1 + p))^\lambda, & x \geq \frac{1+p}{2}. \end{cases}$$

Since $G(x)$ only measures the number of active fans at $x = 1$, we only consider $x \geq (1 + p)/2$, giving

$$(3.12) \quad G(x) = (2x - 1 - p)^{-\lambda} \int (2x - 1 - p)^\lambda \left(\frac{((1 - p)(x^{k-1} - x) - (1 + p)) G'(0) + (1 - p)(x + x^{k-1}) G'(1)}{2x - 1 - p} \right) dx,$$

where integrating, having using a substitution of $u = 2x - 1 - p$, gives

$$(3.13) \quad G(x) = \frac{1}{2} \left((G'(0) + G'(1))(1 - p) \sum_{j=0}^{k-1} \binom{k-1}{j} \left(\frac{1}{2}\right)^j \left(\frac{1+p}{2}\right)^{k-1-j} \frac{(2x - 1 - p)^j}{j + \lambda} \right. \\ \left. + (G'(1) - G'(0)) \frac{1 - p}{2} \frac{(2x - 1 - p)}{\lambda + 1} + \left((1 - p)G'(1) - (1 + p)G'(0) - (1 - p) \frac{1 + p}{2} G'(0) \right) \frac{1}{\lambda} \right),$$

where no additive constant is included because it will be destroyed by satisfying self consistency. After having differentiated and evaluated at $x = 0$ and $x = 1$, (3.14) is obtained, which is the general equation λ must satisfy, true for all k . This is

$$(3.14) \quad \left(2\lambda + 1 + p - 2(\lambda + 1)(1 - p) \sum_{j=0}^{k-1} \binom{k-1}{j} \left(\frac{1}{2}\right)^j \left(\frac{1+p}{2}\right)^{k-1-j} \frac{j(1-p)^{j-1}}{j + \lambda} \right) \cdot \\ \cdot \left(2\lambda + 3 - p - 2(\lambda + 1)(1 - p) \sum_{j=0}^{k-1} \binom{k-1}{j} \left(\frac{1}{2}\right)^j \left(\frac{1+p}{2}\right)^{k-1-j} \frac{j(-1-p)^{j-1}}{j + \lambda} \right) = \\ = (1 - p)^2 \left(2(\lambda + 1) \sum_{j=0}^{k-1} \binom{k-1}{j} \left(\frac{1}{2}\right)^j \left(\frac{1+p}{2}\right)^{k-1-j} \frac{j(-1-p)^{j-1}}{j + \lambda} + 1 \right) \cdot \\ \cdot \left(2(\lambda + 1) \sum_{j=0}^{k-1} \binom{k-1}{j} \left(\frac{1}{2}\right)^j \left(\frac{1+p}{2}\right)^{k-1-j} \frac{j(1-p)^{j-1}}{j + \lambda} - 1 \right).$$

This is the equation, for general k , that defines the relationship between the eigenvalue, λ , and p . The solution that predicts the fragmentation transition for any k is

$$(3.15) \quad p^* = \arg(\lambda_{\text{real}}(p) = 0),$$

where λ_{real} is the real solution for λ of (3.14) in terms of p . Evaluating first at $k = 3$, this gives

$$(3.16) \quad \left(2\lambda + 1 + p - (\lambda + 1)(1 - p) \left(\frac{1+p}{\lambda+1} + \frac{1-p}{(2+\lambda)} \right) \right) \left(2\lambda + 3 + p - (\lambda + 1)(1 - p) \left(\frac{1+p}{\lambda+1} - \frac{1+p}{(2+\lambda)} \right) \right) = \\ = (1-p)^2 \left((\lambda + 1) \left(\frac{1+p}{\lambda+1} + \frac{-1-p}{2(2+\lambda)} \right) + 1 \right) \left((\lambda + 1) \left(\frac{1+p}{(\lambda+1)} + \frac{1-p}{(2+\lambda)} \right) - 1 \right)$$

Solving this numerically, and selecting only the real solutions then gives, in this case,

$$p^* = 0.314596213.$$

From now, k is just an arbitrary parameter, where the dimensions of the Jacobian for the stability have been converted to the order of a polynomial. To compare results, we ran simulations of the adaptive voter model. This was achieved with the Largenet2 library [76] in C++. In this library, the rewiring procedure was designed with directed edges. Fig. 3.1 shows simulated values against generating function analytic values for q -fans, showing agreement with the results of [10]. In this chapter, we arrived at a fragmentation transition point predicted by the

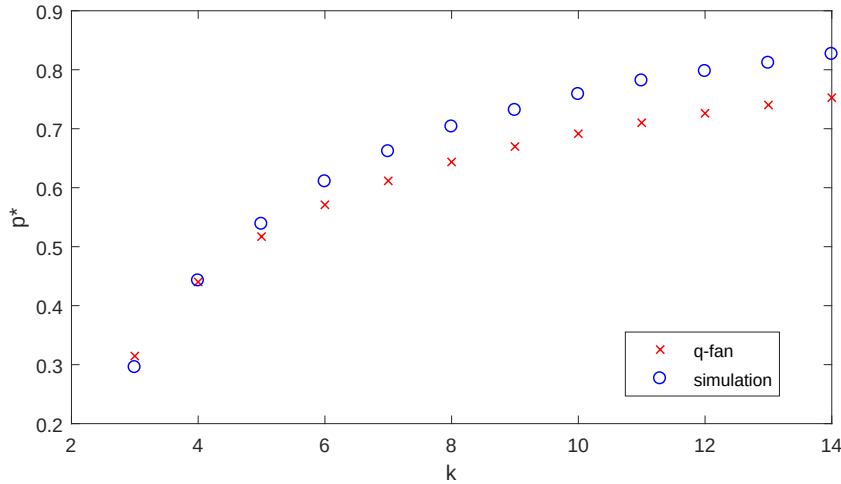


FIGURE 3.1. The plot of the solutions, p^* , of (3.14) for $k \in \{3, \dots, 14\}$, shown as red crosses, against values obtained from simulation, shown as blue circles. Simulations were performed on a network of size 10^4 . This follows exactly the solutions obtained in [10], closely estimating for low k , quickly underestimating for larger k .

active motif q -fan approximation scheme under a regularity assumption. This was found by conditioning on the segregated state, and then analysing the stability of the eigenfunctions of the associated generating function. This compares to the fragmentation point predicted by the active neighbourhood approximation, found in [63], which overestimates for low k but is accurate for high k . The main source of inaccuracy in the regular q -fan approximation is the regularity assumption. Completing the corresponding triple jump analysis demonstrates solvability of this moment approximation, and opens the way to analysing networks with broad degree distributions.

The next chapter applies triple jump methods to analyse the accuracy of an estimate of the epidemic threshold of the susceptible-infected-susceptible (SIS) model.

ERROR IN EPIDEMIC THRESHOLDS FOR THE SIS MODEL

The SIS model [33] is an epidemiological model, used to describe the spread of disease in a static network where no immunity is conferred. It is well understood for mean field approximations, however it can be modelled more accurately with the active neighbourhood approximation. This is less well understood, and leads to a two-dimensional nonlinear PDE, which is likely unsolvable. However, this PDE can be used to identify the source of the error in the simpler approximations.

In the SIS model nodes can be either susceptible S , or infected I . In random time, infected nodes can either infect susceptible nodes with a probability, p , or recover with a probability, r . Infection occurs along $S-I$ links, whereas recovery happens independent of a node's neighbours. These two actions have a nonlinear relationship regarding the long-term behaviour of an SIS model. With high enough p with respect to r , infected nodes may persist within the network for all time. This state is known as an epidemic. The alternative is that the infected nodes are eliminated, which we shall call the cured state.

The susceptible-infected-recovered (SIR) model [38] acts similarly to the SIS. In this case, upon recovery, nodes become immune to the disease, and cannot be reinfected. Its construction means an analytic solution to the epidemic threshold has already been found by using a birth-death process representation. However, no true analytic solution for the epidemic threshold of the SIS model has been found, but there are partial solutions [14, 69]. Following on from the notation and calculations of [62], we analyse the accuracy of such an approximation.

Using the active neighbourhood approximation, denote $S_{k,l}$ as the number of S agents with k S and l I neighbours, and $I_{k,l}$ similarly for I agents. Denote the zeroth order moments as

$$(4.1) \quad S = \sum_{k,l} S_{k,l}; \quad I = \sum_{k,l} I_{k,l},$$

where these are the total densities of S and I nodes in the network, respectively. Since nodes are

conserved, we have that $S + I = 1$. The first order moments are defined as

$$S_S = \sum_{k,l} k S_{k,l}; \quad S_I = \sum_{k,l} l S_{k,l}; \quad I_S = \sum_{k,l} k I_{k,l}; \quad I_I = \sum_{k,l} l I_{k,l},$$

where, say, S_I is the density of S-l links. These motifs are ordered, so that S-S and l-l pairs are double counted. The SIS model conserves links, so that for a network with mean degree z , $S_S + S_I + I_S + I_I = z$. Second order moments are similarly defined, such that

$$\begin{aligned} S_{SS} &= \sum_{k,l} k(k-1) S_{k,l}; & S_{SI} &= \sum_{k,l} k l S_{k,l}; & S_{II} &= \sum_{k,l} l(l-1) S_{k,l}; \\ I_{SS} &= \sum_{k,l} k(k-1) I_{k,l}; & I_{SI} &= \sum_{k,l} k l I_{k,l}; & I_{II} &= \sum_{k,l} l(l-1) I_{k,l}, \end{aligned}$$

where, say, S_{SI} is the density of ordered S-S-l triplets, so triplets are double counted. The original rate equations [62] consider the general adaptive SIS model. This is a variant of the SIS model with a rewiring action, w . Setting $w = 0$ returns the static SIS model, so the rate equations are a pair of coupled infinite-dimensional systems of ODEs, such that

$$\begin{aligned} (4.2) \quad \frac{\partial S_{k,l}}{\partial t} &= -p l S_{k,l} + p \frac{S_{SI}}{S_S} ((k+1) S_{k+1,l-1} - k S_{k,l}) + r I_{k,l} + r ((l+1) S_{k-1,l+1} - l S_{k,l}); \\ \frac{\partial I_{k,l}}{\partial t} &= p l S_{k,l} + p \left(1 + \frac{S_{II}}{S_I}\right) ((k+1) I_{k+1,l-1} - k I_{k,l}) - r I_{k,l} + r ((l+1) I_{k-1,l+1} - l I_{k,l}). \end{aligned}$$

Denote the generating functions with active motif coefficients as

$$(4.3) \quad G^S(t, x, y) = \sum_{k,l} S_{k,l}(t) x^k y^l; \quad G^I(t, x, y) = \sum_{k,l} I_{k,l}(t) x^k y^l.$$

A bar will be used to denote evaluation at $x = y = 1$, such that, say, $\overline{G_{xy}^S} = G_{xy}^S(t, 1, 1)$. Denote $S_{S^m I^n}$ as the $(m+n)^{\text{th}}$ moment, counting the number of motifs with a focal node S, m fringe S nodes, and n fringe l nodes. Denote similarly for $I_{S^m I^n}$. Using generating function, we have that

$$(4.4) \quad \overline{G_{x^n y^m}^S} = S_{S^m I^n}; \quad \overline{G_{x^n y^m}^I} = I_{S^m I^n}.$$

This also means

$$(4.5) \quad \overline{G^S} + \overline{G^I} = 1; \quad \overline{G_x^S} + \overline{G_y^S} + \overline{G_x^I} + \overline{G_y^I} = z.$$

In fact, all moments follow a conservation law. Inserting the generating function definitions (4.3) into the rate equations, (4.2) gives

$$\begin{aligned} (4.6) \quad G_t^S &= p \alpha (y-x) G_x^S + (r(x-y) - p y) G_y^S + r G^I; \\ G_t^I &= p (\beta + 1) (y-x) G_x^I + p y G_y^S + r(x-y) G_y^I - r G^I, \end{aligned}$$

where

$$\alpha = \frac{\overline{G_{xy}^S}}{\overline{G_x^S}}; \quad \beta = \frac{\overline{G_{yy}^S}}{\overline{G_y^S}}.$$

The epidemic threshold is the border of the manifold of steady states of $\overline{G^S}$ and $\overline{G^I}$ that minimise $p(r)$, where $\overline{G^I} > 0$. Due to the conservation law in (4.5), these are the same for $\overline{G^S}$ and $\overline{G^I}$.

To find the correct solution, these equations must be self consistent. Solving for when $I = 0$, the eradication state, will demonstrate this. In this case, $G^I = 0$, and for all $l \geq 1$, $S_{k,l} = 0$. Since there are no l nodes, the network is inactive. This means there is no time dependency, allowing t to be dropped. Then $G^S(t, x, y) = G^S(x) = \sum_k S_{k,0} x^k$. This means all derivatives with respect to y become 0. Hence, (4.6) resolves to $0 = 0$, and so the equations are self-consistent.

A pair of steady state equations can be obtained for the general case by setting $G_t^S = G_t^I = 0$ in (4.6). After rearranging, this gives a pair of steady state equations for G^I , such that

$$(4.7) \quad G^I = \frac{1}{r} \left(p \alpha (x - y) G_x^S + (p y - r(x - y)) G_y^S \right);$$

$$(4.8) \quad G^I = \frac{1}{r} \left(p (\beta + 1) (y - x) G_x^I + p y G_y^S + r(x - y) G_y^I \right).$$

To quantify the error of the mean field approximation, a heterogeneous second order expansion is used, which leads to a two-dimensional PDE.

Say that the l nodes are distributed randomly across a given network. To emphasise that it is now treated as a parameter, denote $I = \rho$. Given a degree distribution where any node has probability p_k of having k neighbours, that means that, by the law of total probability,

$$(4.9) \quad \rho(t) = \sum_k \rho_k(t) p_k(t),$$

where ρ_k is the probability a node is infected, given it has degree k . Using a mean field approximation, as in Pastor-Satorras and Vespignani [54], we get the rate equation,

$$(4.10) \quad \frac{\partial \rho_k(t)}{\partial t} = -r \rho_k(t) + p k (1 - \rho_k) \rho.$$

The destruction term is due to the recovery rate, r . Ignoring any correlations, the creation term is the product of the infection rate, p , the number of neighbours, k , the probability of selecting a susceptible node, given it is of degree k , and the probability any neighbour is infected. In the steady state, where $\partial_t \rho_k(t) = 0$, (4.10) becomes

$$(4.11) \quad \rho_k = \frac{k p \rho}{r + k p \rho},$$

as in where the time dependency is dropped, since this is for asymptotic time. Continuing through, suppose that the active neighbourhood expansions can be approximated via the mean field approximation. This gives

$$(4.12) \quad S_{k,l} = \binom{l+k}{k} \frac{r}{r + (l+k)p\rho} (1-\rho)^k \rho^l; \quad I_{k,l} = \binom{l+k}{k} \frac{(l+k)p\rho}{r + (l+k)p\rho} (1-\rho)^k \rho^l.$$

The next step is to insert these values into the steady state equations, however, the mean field ansatz is of lower order than the active neighbourhood, and so inserting these values will result in an incorrect equation. The goal then is to produce a measure of this error.

Generating functions are polynomials, where the k^{th} coefficient represents some measure of the k^{th} order object in a system. It is also known that for any pair of polynomials, such that for coefficients, $a_{k,l}$, $b_{k,l}$, and variables, x, y , if

$$(4.13) \quad \sum_{k,l} a_{k,l} x^k y^l = \sum_{k,l} b_{k,l} x^k y^l,$$

then for each k, l , $a_{k,l} = b_{k,l}$. Each order of $x^k y^l$ can be considered a separate dimension, orthogonal to any other $x^{k'} y^{l'}$. Therefore, given two generating function expressions in x and y , their difference can be uniquely defined by the difference between each $(k, l)^{\text{th}}$ coefficient. Upon inserting (4.12) into either of (4.7) and (4.8), the RHS can be rearranged into the general polynomial form, so the differences between each coefficient can be read off. This gives a measure of the inaccuracy of the mean field approximation for the number of each active neighbourhood type.

Both (4.7) and (4.8) are similarly inaccurate, as they are equal and because of the conservation laws. Therefore, we need only consider (4.7), subject to (4.12) from this point on.

Applying (4.12), and rearranging the RHS of (4.7) into general polynomial form, we obtain

$$(4.14) \quad \begin{aligned} & \sum_{k=0, l=0} \binom{l+k}{k} \frac{p(l+k)\rho}{r+p(l+k)\rho} (1-\rho)^k \rho^l x^k y^l = \\ & = \sum_{k=0, l=0} \binom{l+k}{k} \frac{p(l+k)\rho}{r+p(l+k)\rho} (1-\rho)^k \rho^l \frac{1}{l+k} \left(\frac{1}{\rho} \left(\alpha k + \left(1 + \frac{r}{p} \right) l \right) - \frac{1-\rho}{\rho^2} \alpha l - \frac{rk}{p(1-\rho)} \right) x^k y^l, \end{aligned}$$

where $\overline{G_x^S}$ and $\overline{G_{xy}^S}$ here are defined according to the mean field ansatz. A termwise error relation can now be obtained by stripping away the summation and $x^k y^l$ parts, giving the error function,

$$(4.15) \quad E(k, l) = \binom{l+k}{k} \frac{l+k}{r+p(l+k)\rho} (1-\rho)^k \rho^{l-1} \left(\rho \left(p - \frac{1}{l+k} \left(k \left(\alpha p + \frac{r}{1-\rho} \right) + l(p-r) \right) \right) + \frac{l}{l+k} \alpha p (1-\rho) \right).$$

This equation should be consistent for both epidemic and eradicated steady states. It is known through simulations that the mean field approximation accurately predicts the trivial steady state, and so must produce no error in that case. The goal then is to prove that, for all k, l , as ρ goes to zero, so does $E(k, l)$. Beginning with $l = 0$, we get that

$$(4.16) \quad E(k, 0) = \frac{(1-\rho)^k k}{r + pk\rho} \left(\rho \left(p - \frac{r}{1-\rho} \right) - p \right).$$

Since the number of infected nodes limits and affects the number of connections to infected nodes, $\overline{G_{xy}^S}$ scales with ρ , while $\overline{G_x^S}$ scales inverse to it. This means that

$$(4.17) \quad \alpha \xrightarrow{\rho \rightarrow 0} 0.$$

The equations support this. The numerator approaches zero because

$$(4.18) \quad \overline{G_{xy}^S} = S_{SI} = \sum_{k=0, l=0} kl \binom{l+k}{k} \left(1 - \frac{p(l+k)\rho}{r+p(l+k)\rho} \right) (1-\rho)^k \rho^l$$

and so all nonzero terms are multiplied by powers of ρ . The denominator stays nonzero because

$$(4.19) \quad \overline{G_x^S} = S_S = \sum_{k=0, l=0} k \binom{l+k}{k} \left(1 - \frac{p(l+k)\rho}{r + p(l+k)\rho}\right) (1-\rho)^k \rho^l,$$

so the $l = 0$ terms in the series are zero only at $\rho = 1$. In (4.15), for $l \geq 1$, the terms in the first pair of brackets have a positive order coefficient of ρ , and are all finite, so approach 0. The second term goes to zero for all integer $l \geq 1$ because we have (4.17). This means that for all k, l ,

$$(4.20) \quad E(k, l) \xrightarrow{\rho \rightarrow 0} 0.$$

The mean field approximation gives no consideration to the shape of motifs. When $\rho > 0$, for all $i > j$, the i^{th} moment is larger than the j^{th} moment. Under these assumptions, that means α is infinite. Assume then that nodes above a certain degree, $d_{\max}$ do not occur. Then,

$$(4.21) \quad \begin{aligned} \overline{G_x^S} &= \sum_{0 \leq k+l \leq d_{\max}} k \binom{l+k}{k} \left(1 - \frac{p(l+k)\rho}{r + p(l+k)\rho}\right) (1-\rho)^k \rho^l; \\ \overline{G_{xy}^S} &= \sum_{0 \leq k+l \leq d_{\max}} kl \binom{l+k}{k} \left(1 - \frac{p(l+k)\rho}{r + p(l+k)\rho}\right) (1-\rho)^k \rho^l, \end{aligned}$$

which are both finite, and so α is a function of $d_{\max}$. The next step is to plug in values of p , r , and ρ for E that are can give epidemic steady states. Using the mean field ansatz, we have that

$$(4.22) \quad \frac{\partial \rho}{\partial t} = pSIz - rI = p(1-\rho)\rho z - r\rho,$$

so that in the steady state, where $\partial_t \rho(t) = 0$,

$$(4.23) \quad \rho = \frac{r}{pz} - 1.$$

It is also known that $r = zp$ is a steady state. However, these conditions together result in degeneracy, so ρ will be chosen to deviate from the solution. Errors under these conditions will inform us about steady states errors, as they are absorbing. Using the mean field, this means $\rho = 0$, so for the nontrivial steady state, $r > zp$. Assume the graph is Erdős Rényi, and so Poisson distributed. With any choice of z , we can select then the maximum degree, $d_{\max}$, such that

$$(4.24) \quad d_{\max} = \arg \left(\min_d \left(\sum_{k=0}^d e^z \frac{z^k}{k!} \geq 0.99995 \right) \right),$$

and $\alpha(d_{\max}) = \alpha(z)$. Plugging $r = zp$ and $\rho = (r/pz) - 1$ into (4.15) gives 0 error, as this is the trivial steady state. Say, for example, $r = 1.1zp$, a close value that is nontrivial. If $z = 4$, then $d_{\max} = 14$. Since $r \leq 1$, $p \leq 5/22$. Say that $p = 0.2$, then $r = 0.88$ and $\rho = 0.1$. Plotted against k, l , this gives Fig. 4.1.

This shows that the mean field assumption gives poor estimates for the number of l nodes with many l neighbours and few S neighbours. These can be thought of as members of local infected

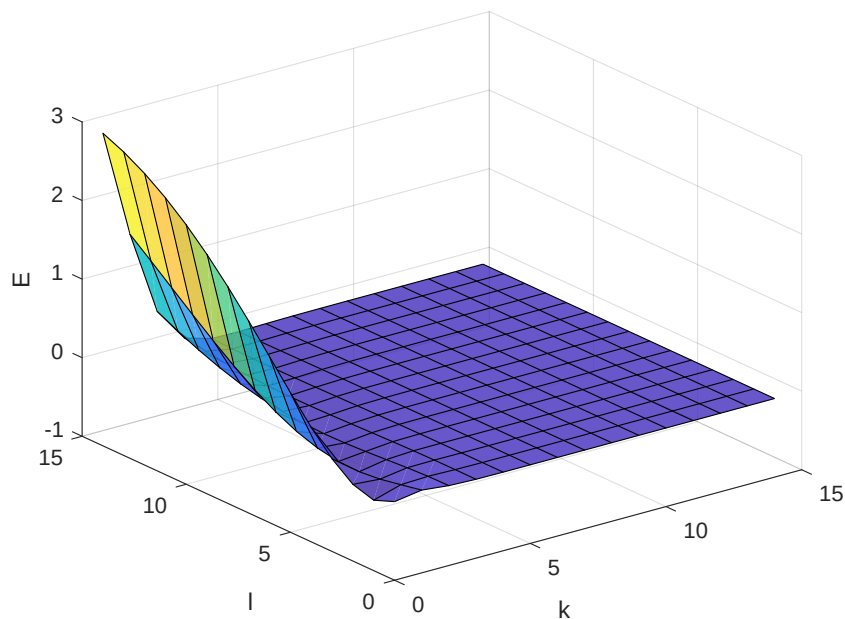


FIGURE 4.1. Plot of the Error from the true number of $I_{k,l}$ spiders, against k and l . Parameters used are $z = 4$, $d_{\max} = 12$, $p = 0.2$, $r = 0.88$, $\rho = 0.1$. One can see that higher l with lower k result in overall poorer estimates, with small underestimation for small l with small k . For higher l , reduction in k accelerates increase in accuracy. For small k , increase in l overall decelerates increase in inaccuracy.

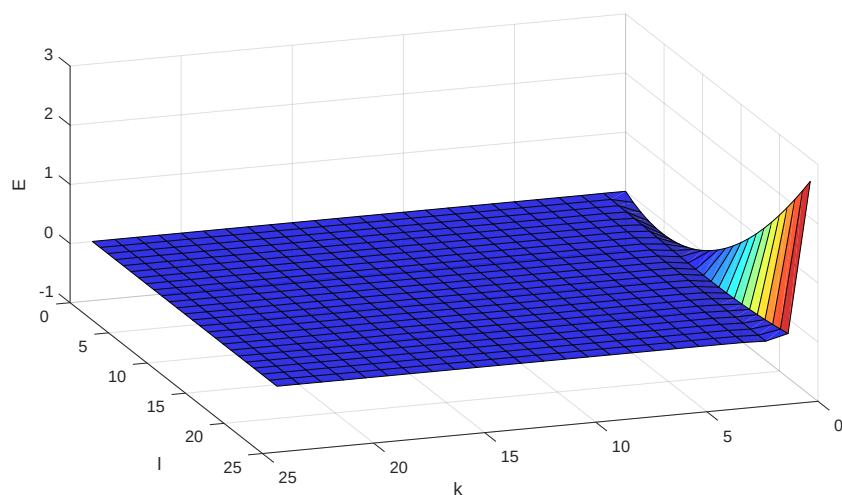


FIGURE 4.2. Plot of the Error from the true number of $I_{k,l}$ spiders, against k and l . Parameters used are $z = 10$, $d_{\max} = 24$, $p = 0.05$, $r = 0.505$, $\rho = 0.01$. One can see a similar relation between E , k , and l , as in Fig. 4.1, but more clearly the sharp inaccuracy for $k = 1$.

clusters who are one infection away from being fully embedded. That this sharp inaccuracy is for single k but all l , increasing as embedding does, is shown more clearly in Fig. 4.2.

With Fig. 4.1 and Fig. 4.2, and the smooth progression that can be expected from the restrictions on the parameters, it can be seen that l nodes one infection away from full embedding in high degree clusters are highly overestimated. However, as shown in Fig. 4.3, this is only for l nodes whose number of l neighbours is larger than z . Otherwise the mean field assumption gives an underestimate.

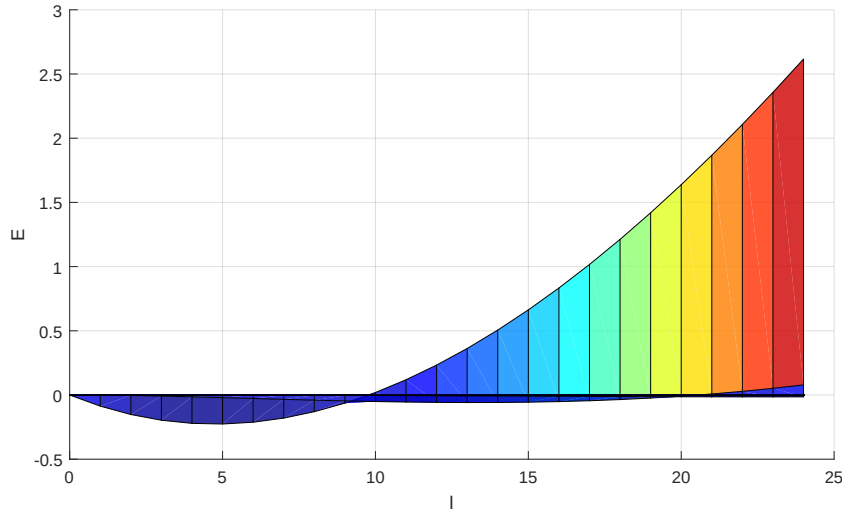


FIGURE 4.3. The side view of E against l of a plot of the Error from the true number of $I_{k,l}$ spiders, against k and l . Parameters used are $z = 10$, $d_{\max} = 24$, $p = 0.05$, $r = 0.505$, $\rho = 0.01$. Most values below $l = 10$ are underestimates that return to 0, while most above are overestimates.

For all other k, l , the mean field assumption closely estimates the true rate of occurrence of spider motifs. Since the degree of overestimation positively correlates with embedding of l nodes with an S neighbour above the mean, it overestimates the number of infectious nodes that can be expected to survive indefinitely. In addition, the longer they are expected to survive, the greater the overestimate. The mean field approximation is then expected to underestimate the epidemic threshold.

The next chapter looks at the correspondence between network dynamics and generating functions from a constructive perspective, to produce general PDEs for different networks. As a base case, the single state network without correlations is studied.

CONSTRUCTING NETWORKS FROM PDES

The number of models that can be solved completely will always be limited. Networks are also not fully understood, whereas the solution space of PDEs is well explored. Generating functions map network dynamics into PDEs, and so using this framework one can map out the space of network models that can be solved easily.

For any enumerable set of objects, assigning them to a generating function is an isomorphic mapping. This means that no information is lost, and so one may similarly produce any enumerable set from a given generating function. As we have seen, the degree distributions, or normalised degree counts, of a network may be represented in the form of a generating function. This isomorphic mapping means one may instead generate networks and their dynamics by a choice of generating function.

To create a general PDE for the dynamics of a generating function, the PDEs for each possible action must first be identified. This is done by analysing the agent based dynamics, and then adding the associated PDEs together, with rate parameters for each independently occurring action. From this, the dynamics of any model that has these behaviours can be generated by a choice of parameters in this PDE. If we say for some generating function,

$$G(x; t) = \sum_k p_k(t) x^k,$$

where p_k is a number of nodes with degree k , and we, as before, drop the time dependence unless it becomes explicitly important. First, let us examine some trivial cases. These will be describing static networks. Say first that $G = 0$. This corresponds to a null graph, because there are no nodes of any degree. An order above this is $G = c$, where c is some constant. This gives an empty graph, with disjoint nodes. In this case, either $c = 1$ or some integer, depending on whether we normalise or not. Finally, the general form of any static network is $G = c(x)$, where c in this case is some function of x . For example, the Erdős-Rényi random graph with mean degree z would have the

generating function, $G = e^{z(x-1)}$. This is because the graph has a Poissonian degree distribution, and so G is exactly the probability generating function of a Poisson distribution.

Generating function results for static graphs are well established, but introducing dynamic behaviour moves us away from one dimensional system, adding differential equations. The choice of moment approximation and motif determines the order of the differential system. Since generating function mappings are isomorphic, the converse is also true. The next step would then be, for some function F , to study a first order differential equation of G , with general form $dG/dt = F(x, t, G, G_x)$.

To examine Markovian systems, no equations with t dependencies are considered. The following processes are considered in terms of the heterogeneous mean field approximation, and for a network with only one node state. Given this, the only possible actions result in dynamics of the network. The simplest operations available are node addition and removal, and link addition or removal, while any other definable action is some combination or subset of these. Hence, the dynamic equation associated with any definable action will be a linear combination of those associated with link addition and removal, and node addition and removal, as will its solution.

The following is an examination of what could be considered basic network actions, and rate parameters are omitted in the absence of any more than one independent process.

Say we consider a network with spontaneous random link removal. We are using the mean field, and study this as a node update action. For this, we have that, for any node of degree k ,

$$(5.1) \quad \frac{dp_k}{dt} = (k+1)p_{k+1} - kp_k + z(q_k - q_{k-1}) = 2(k+1)p_{k+1} - 2kp_k,$$

Link removal decreases the order of their connected nodes by one. Any link always has kp_k chances to be on a node of order k , by randomly selecting a node of order k , and then having k chances for it to have our given link. Any link also affects excess nodes similarly, giving (5.1). Link addition is somewhat different because, upon selecting a root node, the node one wires it to may neither be the original, nor its neighbourhood. It is also not defined by the excess degree distribution, so under the mean field ansatz it is otherwise independent. However, this independence results in a similar solution to (5.1). That is, for link addition,

$$(5.2) \quad \frac{dp_k}{dt} = -p_k + p_{k-1} - \left(p_k - \frac{1+z}{\sum_i p_i}\right) + \left(p_{k+1} - \frac{1+z}{\sum_i p_i}\right) = 2p_{k-1} - 2p_k,$$

since this action increases order, while still selecting from the complementary, and independent event space. The two link actions are visualised in Fig 5.1.

What this means in terms of generating functions is that, for link removal,

$$\frac{dG}{dt} = \sum_k (2(k+1)p_{k+1} - 2kp_k)x^k = 2\sum_k kp_k x^{k-1} - 2x \sum_k kp_k x^{k-1} = -2(x-1)G_x.$$

For link addition, this gives

$$\frac{dG}{dt} = \sum_k (2p_{k-1} - 2p_k)x^k = 2x \sum_k p_k x^k - 2 \sum_k p_k x^k = 2(x-1)G$$

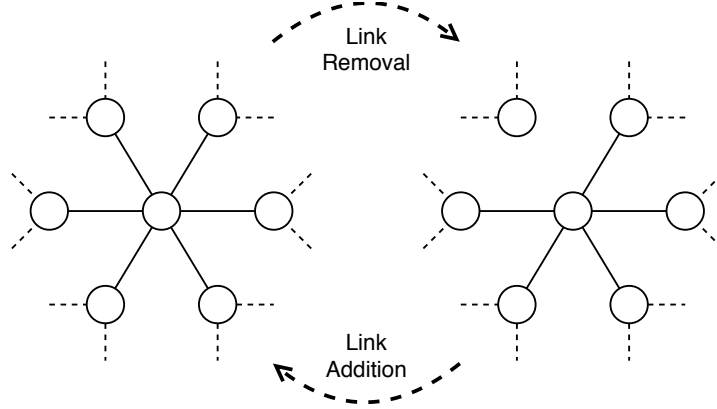


FIGURE 5.1. The action of link removal and addition on some arbitrary single state network.

Consider now node actions. The issue with this is that nodes are coupled with links, so node addition or removal also constitutes link addition or removal, unless we restrict ourselves to only adding or removing isolated nodes.

This brings the question of how to add or remove a node. For removal, the most obvious choice is randomly deleting nodes with their incident edges, and we shall call this simple node removal. In simple node removal, the network structure has implicitly dictated that this deletes a node, and reduces the degree of its neighbours. For addition, network structure can do no such thing, since the node does not yet exist. The way then is to select an already present node, call it a , create a new node, b , and wire it to all the neighbours of a , copying it. We call this duplication. Both behaviours are shown in Fig. 5.2.

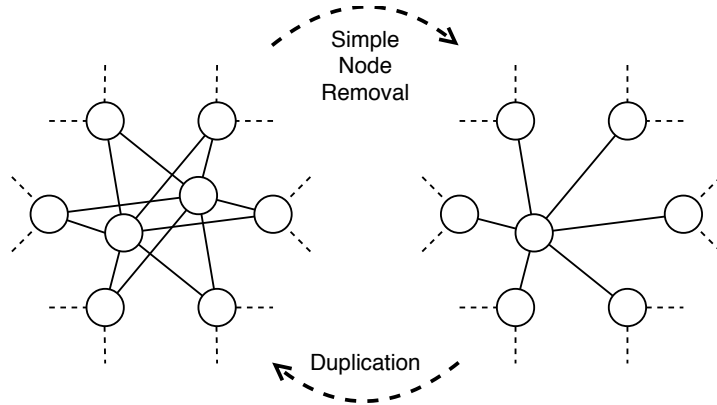


FIGURE 5.2. The action of simple node removal and duplication on some arbitrary single state network.

With these actions, the rate equation for simple node removal is

$$(5.3) \quad \frac{dp_k}{dt} = -p_k + z(q_k - q_{k-1}) = (k+1)(p_{k+1} - p_k)$$

where z is the mean degree, and q_k is the number of nodes with k excess neighbours. The p_k term is because this removes one node of degree k with p_k many chances to do it. The excess terms describe the decrease in degree of neighbouring nodes, of which there are on average z .

On the other hand, the rate equation for duplication is

$$(5.4) \quad \frac{dp_k}{dt} = p_k + z(q_{k-2} - q_{k-1}) = p_k + (k-1)p_{k-1} - kp_k = (k-1)(p_{k-1} - p_k),$$

which can be seen to be the reverse of simple node removal. We now construct their generating function rate equations. For simple node removal, this is

$$\frac{dG}{dt} = \sum_k (k+1)(p_{k+1} - p_k)x^k = \sum_k kp_k x^{k-1} - x \sum_k kp_k x^{k-1} - \sum_k p_k x^k = -(x-1)G_x - G,$$

while for duplication it is

$$(5.5) \quad \begin{aligned} \frac{dG}{dt} &= \sum_k (k-1)(p_{k-1} - p_k)x^k \\ &= x \sum_k kp_k x^k - x \sum_k kp_k x^{k-1} + x \sum_k p_k x^{k-1} \\ &= (x^2 - x) \sum_k kp_k x^{k-1} + \sum_k p_k x^k \\ &= G + x(x-1)G_x. \end{aligned}$$

The problem with these two actions is that they do not independently remove or add a node, but also alter the degrees of their neighbours. Some removal and addition actions for nodes must be used that only change the number of nodes. Since network structure can simplify a rule implicitly, removal actions shall be first considered.

Since simple node removal only affects the removed node's neighbourhood, a fix need only address the neighbours' degree changes. Their degrees must be preserved, which will be done by adding edges to them, but they are the only nodes that have reduced their degree.

Therefore, they should wire amongst themselves, keeping the network structure intact. For any node with degree k , this means, upon removing it and its edges, wiring $\lfloor k/2 \rfloor$ many edges between k nodes such that no node has degree greater than one on the induced neighbourhood subgraph. We shall call this process removal with rewiring, and is shown in Fig. 5.3.

For odd sized neighbourhoods, this always leaves one node without a neighbour. A solution is then to randomly choose between $\lfloor k/2 \rfloor$ and $\lfloor k/2 \rfloor + 1$ many edges to wire, which, in a random process, approximates to $k/2$, and so gives

$$(5.6) \quad \frac{dp_k}{dt} = -p_k.$$

To preserve simplicity, this depends on there being few enough triangles in the network to allow for sparse induced subgraphs on any neighbourhood of a node. On most infinite networks, this is not a problem.

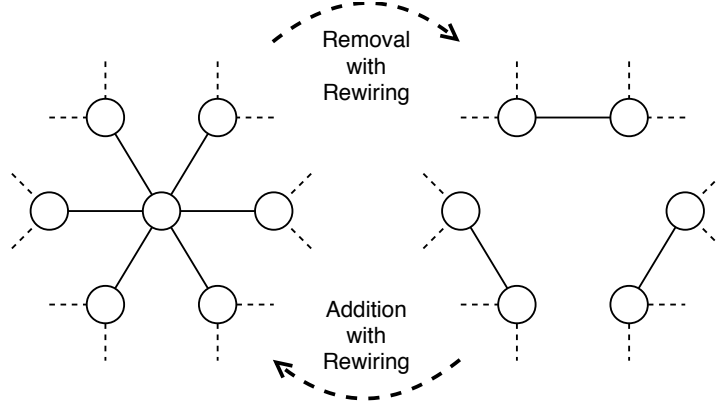


FIGURE 5.3. The action of removal with rewiring and addition with rewiring on some arbitrary single state network.

Onto addition. The simplest way to add a node without affecting the degrees of any others is to make it isolated. Denoting $\delta_{i,j}$ as the Kronecker delta, spontaneously spawning isolated nodes therefore gives the change equation,

$$(5.7) \quad \frac{dp_k}{dt} = \delta_{0,k} r.$$

This clearly has no system dependency, as it is an isolated process. Note that this action is an element of the set of actions that spontaneously add nodes of fixed degree, say m . What makes it unique is that one need not consider how to wire the links of the generated node to the network, as there are none. For any $m \geq 1$, this must be specified. For this action set, the simplest randomly assigns link through the network, giving

$$(5.8) \quad \frac{dp_k}{dt} = \delta_{m,k} r + rm(p_{k-1} - p_k).$$

Spontaneous isolated node spawning is not a reversal of removal with rewiring, since there is no link behaviour. The added node must alter the degree distribution randomly. Since addition has no implicit behaviours dictated by the network structure, the addition rule must again have more conditions than the removal rule.

Defining stubs as the end of an edge touching a node, first select a random node, n_0 , on the graph. Then randomly select a set of nodes, call it s_0 , such that $|s_0| = \deg(n_0)$ and the number of stubs is greater than $|s_0|$. Generate a new node, n_1 , and then wire every node of s_0 to n_1 , using the already present stubs. Connect leftover stubs in between the nodes of s_0 until all stubs are used. If there is still a leftover stub, randomly discard or wire it to a node that would preserve simplicity. This step approximately preserves the degrees of all nodes. Then a new node is created according to the present degree distribution, that preserves the degrees of all nodes in the network, such that

$$(5.9) \quad \frac{dp_k}{dt} = p_k.$$

This is a reverse of removal with rewiring, as shown by (5.9). The generating function rate equation for removal with rewiring is

$$(5.10) \quad \frac{dG}{dt} = -G,$$

The solution to this is $G = Ae^{-t}$, where A is some constant. This is because the edge structure, at the zeroth degree, is never changed. Thus, using zeroth degree moments results in the same process as random particle decay, which is the basic Poisson process. In reverse, the generating function rate equation for addition with rewiring is

$$(5.11) \quad \frac{dG}{dt} = G.$$

This sets out a complete set of basic actions to construct the dynamics of any general dynamic network with only one node type. Say then there is a network where links are removed with rate $r_1/2$ and added with with rate $r_2/2$, while nodes are removed with rewiring at a rate of r_3 and added with rewiring at a rate of r_4 , where all actions occur independently. Then the rate equation for this network is

$$(5.12) \quad \frac{dG}{dt} = r_1(1-x)G_x + (r_4 - r_3 - r_2(1-x))G,$$

which is a 1st order, linear homogeneous PDE, and therefore solvable. For any arbitrary function, f , (5.12) has the general solution,

$$(5.13) \quad G(x, t) = f\left(t - \ln\left((x-1)^{\frac{1}{r_1}}\right)\right)(x-1)^{\frac{r_3-r_4}{r_1}} e^{\frac{r_2}{r_1}x}.$$

This means that any single state network with spontaneous dynamics can be created by a choice of parameters in (5.13), and its dynamic behaviour as well as degree distribution is fully known. Similarly, one may choose an expression for $G(x, t)$ to satisfy, giving the rates of the basic actions in a single state spontaneous model. For example, insert $G(x, t) = e^x$ into (5.13). This is solved by $f = 1$, $r_3 = r_4$ and $r_2 = r_1$, giving a stationary dynamic network, where nodes and links are lost as fast as they are gained. Alternatively, say $G(x, t) = e^{z(x-1)}$, the Poisson probability generating function. Then, (5.13) is satisfied by $f = e^{-z}$, $r_3 = r_4$, and $r_2 = zr_1$. That is, a stationary network where links are removed as fast as they are added, and nodes are added z faster than they are removed, with a normalisation factor of e^{-z} .

Aside from obtaining results about the solvability of networks, triple jump analysis has been used in the design of their topology, as in [64]. However this was done to create topologies that generate desired steady state distributions, and did not result in many solvable models. By furthering the work to build up the space of solvable models, alternative insights may be obtained to those found with this approach.

The conclusion in the next chapter discusses the results and limitations present within this project, and speculates about further research.

CONCLUSION AND FUTURE WORK

The triple jump procedure, by representing the dynamics of adaptive networks as a generating function, enables purely analytic study of some network behaviours. In this report, triple jump methods have been employed in different ways to explore the dynamics of the adaptive voter model, the SIS model, and a simple spontaneous model.

In Chapter 3, the dynamics of the adaptive voter model were captured using an active motif approximation, conditioned on segregation, assuming degree regularity. With the triple jump method, this extended the work of Böhme and Gross [10] by reducing the infinite-dimensional system of equations into a one dimensional PDE, and solving for its stability. This gives an analytic solution to the fragmentation transition, in line with the original paper. However, the matrix approach cannot generalise to networks with broad degree distributions, because this would give infinitely large matrices. By demonstrating solvability, our result prepares for future work that removes the degree regular assumption, which is the primary source of error, to generalise to distribution determined dynamics, and solve with the triple jump.

In Chapter 4, the dynamics of the SIS model were captured using the sophisticated active neighbourhood approximation, in an infinite-dimensional system of equations, offering a high degree of accuracy [70]. Using the triple jump method, this infinite system was reduced to a two dimensional PDE [62]. Unable to solve the PDE, we instead used it to find the source of error in the more common mean field approximation. Our analysis showed that, of the infected base motifs, errors are mainly on motifs that are embedded in infected clusters, with one susceptible neighbour. This underestimates for infected degree below the mean, and overestimates above the mean, scaling with degree. There is nearly no error for other motifs. Thus the mean field approximation overestimates active infectious nodes that are expected to persist. This is a valuable insight for developing more accurate approximations in the future. Further work would involve solving the second rate equation to find a solution for the topology of the first moments.

In Chapter 5, a broad class of simple models comprising the elementary processes of network growth and destruction were characterised with the simple mean field approximation, under the triple jump procedure. A general PDE was constructed and solved, such that any equation of that form can create a network model with fully known degree distribution. Further work would be to repeat this analysis for more complex models which can be expected to have solvable PDEs.

This project was limited by the complexity of the PDEs encountered. An issue with generating function PDEs is their dependency on evaluations of the generating function itself, resulting in nonlinearities which may be unsolvable. For example, it was found that using the (m, l) -spiders to evaluate the adaptive voter model resulted in a term that swapped the x and y variables in the dependencies, and hence too complicated to pursue. Included in Section 6 is also an analysis of the steady state solutions to the adaptive voter model approximated by q -fans, as in Chapter 3. It was expected to find fragmentation transitions that agree with those found, however they overestimate, yet have the correct shape. Further research would be to find the cause.

The methodologies developed in this dissertation are not limited to the models we have analysed, as these give analytic results and a method to scrutinise them. As an extension of the adaptive voter model, by adding uncertain states [56] or uninformed states [18], rich dynamics are expected to arise, and characterising them has real physical use. The adaptive SIS [30] is another sensible extension, with multiple different states, as is the variant in Scarpino et al. [59].

Not all network models can be expected to have a general solution, which can be seen in Table 2.1. Whether these or other models correspond to certain PDE subclasses is an important question, which can be answered by using the triple jump procedure.

Alternatively, elementary network actions can be analysed to find associated solvable PDEs. These can be combined to form solvable models with configurable properties. These may include the number of states, if there is copying, spontaneous change, rewiring, actions on disagreement, or a birth and death process. For example, if increasing the number of states, one may study spontaneous state change, which is certainly solvable. After building enough classes of solvable networks, patterns and common characteristics should begin to emerge. For instance, a question to study is which network models lead to elliptical PDEs, as this should lead to new insights.

Network science is still a young field. Most analytics is performed on simple models, some of which are accurately modelled with approximations, but for some this is not yet possible. As the set of uncharacterised models shrinks, and we look to more precisely model physical systems, more complex models will be studied. Then two questions arise; how to create them, and if they are solvable. This dissertation is an early step in that regard. In defining networks by their associated PDEs, solvability is reduced to a question of information flow. By constructing their dynamics from solvable equations, complex models with known topologies can be formed. By being able to analyse the accuracy of their approximation schemes, new ones can be constructed with solvability in mind. The methods discussed in this dissertation can be developed in many clear ways, and provide a basis to answer all three concerns.

Solvability of Linear PDEs

For a linear equation of the second order, with constants, A, B, C, D, E, F , where we denote $\partial^2 u / (\partial x_i \partial x_j) = u_{x_i x_j}$, and the derivatives are well behaved, the general form is

$$A u_{x_1 x_1} + 2B u_{x_1 x_2} + C u_{x_2 x_2} + D u_{x_1} + E u_{x_2} + F u = 0.$$

By taking the PDE discriminant, $B^2 - AC$, we may classify this PDE. This defines the *information flow*. Let lower order terms be denoted by l.o.t. If $B^2 - AC < 0$, then a PDE is elliptical, and can be reduced to

$$u_{x_1 x_1} + u_{x_2 x_2} + \text{l.o.t.} = 0,$$

with an example in the Laplace Equation, $u_{x_1 x_1} + u_{x_2 x_2} = 0$. If $B^2 - AC = 0$, then a PDE is parabolic, and can be reduced to

$$u_{x_1 x_1} + \text{l.o.t.} = 0,$$

with an example in the diffusion equation, $u_t - u_{x_1 x_1} = 0$, where t is time. If $B^2 - AC > 0$, then a PDE is hyperbolic, and can be reduced to

$$u_{x_1 x_1} - u_{x_2 x_2} + \text{l.o.t.} = 0,$$

with an example in the wave equation, $u_{tt} - u_{x_1 x_1} = 0$.

These reductions are known as the canonical forms, and are arrived at through changes of variables. This set of classifications is also true if the PDE is quasilinear, where only the second derivative is linear. The class, boundary conditions, and linearity determine what sort of solution to use. There are many methods for solving second order linear PDEs, with a number of texts, such as Ockendon et al. [53] on the subject. Beyond this, there exist N state linear second order PDEs, of the form

$$\sum_{i,j=1}^N A_{ij} u_{x_i x_j} + \sum_{i=1}^N A_i u_{x_i} + A_0 u = 0.$$

For the matrix $A = (A_{ij})$, it can be proved that, for the change of variables such that

$$\sum_{i,j=1}^N A_{ij} u_{x_i x_j} = \sum_{k=1}^N d_k u_{x_k x_k},$$

the set of d_k are the eigenvalues of A . If the eigenvalues are all positive or all negative, then the PDE is elliptical, with canonical form

$$\sum_{i=1}^N u_{x_i x_i} + \text{l.o.t.} = 0.$$

If one eigenvalue is zero, and the rest are all of the same sign, then the PDE is parabolic with canonical form

$$\sum_{i=2}^N u_{x_i x_i} + \text{l.o.t.} = 0.$$

If no eigenvalues are zero, and all but one eigenvalue are of the same sign, then the PDE is hyperbolic with canonical form

$$u_{x_1 x_1} - \sum_{i=2}^N u_{x_i x_i} + \text{l.o.t.} = 0.$$

Again, different solution schemes exist for all these types of equations, many of which are unsolved.

Steady State Evaluation for the Adaptive Voter Model

The steady state solution for the adaptive voter model is

$$0 = \left(\frac{1-p}{2} (x^{k-1} - x) - \frac{1+p}{2} \right) G_x^*(0) + \frac{1-p}{2} (x + x^{k-1}) G_x^*(1) + \left(\frac{1+p}{2} - x \right) G_x^*(x),$$

which is a first order linear homogeneous ODE. Consider when $x = 1$. This gives

$$(1) \quad G_x^*(1) = \frac{1+p}{1-p} G_x^*(0),$$

and so

$$0 = \left(x^{k-1} + px - \frac{1+p}{2} \right) G_x^*(0) + \left(\frac{1+p}{2} - x \right) G_x^*(x),$$

which, after rearranging, simplifying, and integrating, implies

$$G^*(x) = G_x^*(0) \int_0^x 2 \frac{\xi^{k-1}}{2\xi - (1+p)} + 2p \frac{\xi}{2\xi - (1+p)} - \frac{1+p}{2\xi - (1+p)} d\xi.$$

The next step is to substitute $u = 2\xi - (1+p)$, giving

$$(2) \quad G^*(x) = G_x^*(0) \int_{-(1+p)}^{2x-(1+p)} 2^{2-k} \frac{(u+1+p)^{k-1}}{u} + p - \frac{(1+p)(1-p)}{u} du.$$

By the binomial theorem, it can be seen that

$$\begin{aligned} & \int_{-(1+p)}^{2x-(1+p)} \frac{(u+1+p)^{k-1}}{u} du = \\ (3) \quad & = \int_{-(1+p)}^{2x-(1+p)} \sum_{i=0}^{k-1} \binom{k-1}{i} u^{i-1} (1+p)^{k-1-i} du \\ & = (1+p)^{k-1} \ln \left| \frac{2x-(1+p)}{1+p} \right| + \sum_{i=1}^{k-1} \binom{k-1}{i} (1+p)^{k-1-i} \frac{(2x-(1+p))^i - (-(1+p))^i}{i}. \end{aligned}$$

This means that (2) becomes

$$(4) \quad G^*(x;p) = G_x^*(0) \left(\left(2^{2-k}(1+p)^{k-1} - (1+p)(1-p) \right) \ln \left| \frac{2x - (1+p)}{1+p} \right| + 2^{2-k} \sum_{i=1}^{k-1} \binom{k-1}{i} (1+p)^{k-1-i} \frac{(2x - (1+p))^i - (-(1+p))^i}{i} + 2px \right),$$

and so

$$(5) \quad \begin{aligned} G^*(1;p) &= G_x^*(0) \left(\left(2^{2-k}(1+p)^{k-1} - (1+p)(1-p) \right) \ln \left(\frac{1-p}{1+p} \right) + 2^{2-k} \sum_{i=1}^{k-1} \binom{k-1}{i} (1+p)^{k-1-i} \frac{(1-p)^i - (-(1+p))^i}{i} + 2p \right) \\ &= G_x^*(0) f(p, k), \end{aligned}$$

where $G^*(1;p)$ describes the total number of active fans in the network for a stationary state, and is shown in Fig 1 for $k = 2, \dots, 9$. It is emphasised that this produces an overestimate not in line with simulations, and overestimates the fragmentation transition greatly.

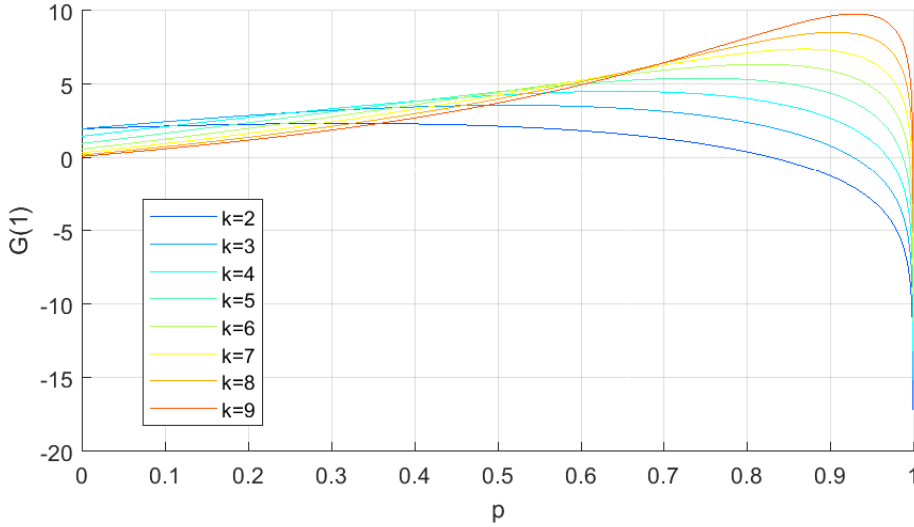


Figure 1: The plot of the estimated number of fans as a function of rewiring rate, for $k = 2, \dots, 9$. $G_x^*(0)$ has been fixed at 1. On all curves the p -intercept tends to the right with k , bounded from the right by a negative asymptote at $p = 1$. $G(1, 0) = 2$ for $k \in \{2, 3\}$, but for all $k \geq 3$ tends to 0 with k . The zeros of this should match those found in [10] for q -fans, but are overestimates.

This has two solutions, either 0, or one that scales with the size of the network, such that as network size approaches infinity, so does the second solution to $G^*(1;p)$. This means that as p increases from 0, $G^*(1;p)$ moves from an unstable infinite solution to a stable 0 solution at p^* in a bifurcation. At this point, both $G_x^*(0)$ and $f(p, k)$ go to 0. Assuming $G_x^*(0)$ is orthogonal to

$f(p, k)$, it should be sufficient to solve for $f(p, k) = 0$, which leads to the equation,

$$(6) \quad 1 = \left(\frac{1-p}{1+p} \right)^{2^{1-k}(1+p)^{k-1} - (1+p)(1-p)^{k-1}} \prod_{i=1}^{k-1} \exp \left(2^{1-k} \binom{k-1}{i} (1+p)^{k-1-i} \frac{(1-p)^i - (-(1+p))^i}{i} + p \right),$$

where $\exp$ is the exponential function. This is finite for $p \in [0, 1]$, and gives the same solutions as $G^*(1; p) = 0$, helping with numerical analysis. The plot for this overestimate is given in Fig 2 for $k \in \{2, \dots, 14\}$ against the simulated estimates for the fragmentation transition. Simulations were run using the Largenet2 library [76].

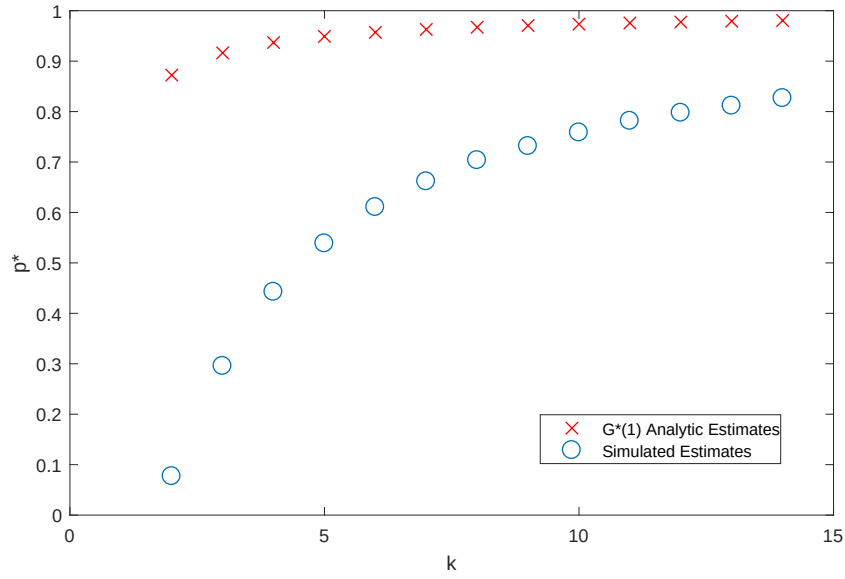


FIGURE 2. The plot of the solutions, p^* , of (6) for $k \in \{2, \dots, 14\}$, shown as red crosses, against values obtained from simulation, shown as blue circles. Simulations were performed on a network of size 10^4 .

Fig. 2 shows that, against simulations, the solutions obtained for 6 are not approximating p^* , giving substantial overestimates. The reason for this could be the subject of future research.

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